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# The impact of monetary policy on the exchange rate: A high frequency exchange rate puzzle in emerging economies



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### A B S T R A C T

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This study investigates the impact of monetary policy shocks on the exchange rates of Brazil, Mexico and Chile. We find that even a focus on 1 day exchange rate changes following policy events – which reduces the potential for reverse causality considerably – fails to lend support for the view that associates unexpected interest rate hikes with immediate appreciations. This lack of empirical backing for the predictions of standard open economy models persists irrespective of whether we use the US Dollar or effective exchange rates, whether changes in the policy rate that were followed by exchange rate interventions are excluded, whether “contaminated” events are dropped from the analysis or whether we allow for non-linearities. We argue that it is difficult to attribute this stronger version of the exchange rate puzzle to fiscal dominance, as unexpected rate increases are not associated with increases in risk premia, and similar results are obtained in the case of Chile – a country that has had the highest possible short-term credit rating since 1995 and a debt/GDP ratio below 10%.

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*“If the relationship between interest rates and exchange rate movements were predictable, the effectiveness of the exchange rate transmission channel would be helpful to monetary policy. But in practice this relationship remains rather inscrutable.”*

Ian Plenderleith, Deputy Governor, South African Reserve Bank

## 1. Introduction

The extent to which monetary factors determine the value of a currency has been one of the key questions of macroeconomics for a long time. While an extensive list of studies has addressed this question and the associated puzzles in the case of developed countries, the literature that looks at this issue in the context of emerging markets has been much more scant, in part because most emerging markets do not have a sufficiently long track record with a floating exchange rate regime. As the number of emerging economies that let the value of their currency be driven by market forces has increased, and the value of the currencies of these increasingly important players in the global economy has come to the forefront of the international policy debate, the need of a clearer understanding of the determinants of the value of their currencies has raised the necessity of such analysis. At the very least, some knowledge about the actual relationship between interest rate and exchange rate movements in emerging economies is required to guide model selection in the context of emerging markets and provide a clearer understanding of the transmission mechanisms that are at work.

Conventional open economy models that combine an uncovered interest parity condition with rational expectations, along the lines of the seminal paper by [Dornbusch \(1976\)](#), would suggest that an unexpected monetary contraction leads to an immediate appreciation of the currency, so as to create the conditions for a subsequent depreciation at a rate that equals the interest rate differential. More generally, the association of interest rate hikes with currency appreciations is also a standard feature of many other workhorse models of international macro and, indeed, of macroeconomic textbooks. Identifying the effects of monetary policy shocks on exchange rates in the data however is not a trivial task if one takes proper account of the issue of endogeneity, i.e. the possibility that monetary policy actions constitute a reaction to concomitant changes in the exchange rate or foreign monetary policy conditions. Indeed, an important study by [Zettelmeyer \(2004\)](#) has argued that the frequent association of positive interest rate shocks with currency depreciations – which is a well known feature of the VAR literature (see for instance [Grilli and Roubini \(1995\)](#) and [Hnatkovska et al. \(2011\)](#)) – can be attributed to this problem of reverse causality.<sup>2</sup> In order to control for this possibility, this paper follows the event study approach taken by Zettelmeyer, to study the impact of monetary policy committee decisions on the exchange rates of the Brazilian Real, the Mexican and the Chilean Peso using daily variations.<sup>3</sup> The main advantage of the event based methodology is that it allows us to properly control for the fact that the monetary authorities of these emerging economies have frequently intervened in the foreign exchange markets – something that has been ignored in earlier developing country studies. Furthermore, a second advantage of our methodology is that our results are not model dependent and, in particular, do not rely on VAR based identification of monetary policy shocks and the strong information assumptions that underlie it. It is important to remember that VAR based procedures require the strong assumption that the information set of the central bank is fully described by the variables contained in the system. The main caveat of our choice is that we are unable to draw any conclusion about the dynamic response of the exchange rate to monetary policy shocks.

The focus on the cases of Brazil, Mexico and Chile allows us to compare the effects of monetary policy in emerging economies that have established floating exchange rate regimes, but have had markedly different levels of gross indebtedness. According to the *de facto* classification of [Reinhart and Rogoff \(2004\)](#) – and later

<sup>2</sup> Hnatkovska, Lahiri and Vegh's study argues that the exchange rate depreciates in response to positive shocks to the interest rate differential in 37 out of 49 developing countries. Their conclusions are based on monthly data.

<sup>3</sup> In contrast to Zettelmeyer, however, we do not restrict our study to monetary policy decisions that led to changes in the policy rate (as decisions to hold may also have a surprise component) and do not include periods in which the central banks responded to exchange rates in an explicit way.

updates of it by the IMF, the first 2 of these countries were the only ones among the 5 largest emerging economies of the world that have operated under a floating exchange rate regime in an uninterrupted fashion for more than a decade.<sup>4</sup> The contribution of our study of 238 monetary policy events is to show that, contrary to the results that were obtained for a number of developed economies,<sup>5</sup> the detailed analysis of the events that surrounded the (pre-scheduled) monetary policy committee meetings that took place in Brazil, Mexico and Chile between January 2003 and May 2011 fails to reverse the exchange rate puzzle, i.e. to provide any evidence that interest rate increases are associated with exchange rate appreciations on impact. Indeed, this contradiction of the textbook relation is very robust and holds irrespective of whether interest rate changes are anticipated or not or of whether changes in the policy rate that were followed by exchange rate intervention are excluded from the sample. Moreover, it persists even if we allow for the possibility of a non-linear relation between interest rates and exchange rates. If anything, we find that the exchange rate depreciates following monetary policy contractions, irrespective of whether we use changes in the US Dollar rate or the variation against a basket of currencies. Second, we also show that the response of the exchange rate to domestic and to US interest rate shocks is clearly not symmetric.<sup>6</sup> While responses to domestic rate changes seem to defy the conventional wisdom, the responses to US rate changes on FOMC meeting days do seem to be in accordance with it. Third, we argue that this puzzling finding cannot easily be attributed to fiscal dominance,<sup>7</sup> as unexpected rate increases are not associated with increases in risk premia in any of these countries and the results hold not only in the case of Brazil or Mexico, but also in the case of Chile – a country that has had the highest possible short-term sovereign credit rating since 1995 and a debt/GDP ratio below 10%. The empirical failure of the conventional view of the exchange rate in these emerging markets represents a higher frequency and more robust version of the exchange rate puzzle that is typical of the VAR literature. While we are able to point out clear evidence of foreign exchange market inefficiencies in the cases of Brazil and Mexico, the asymmetric response to domestic and to US based monetary policy shocks seems to indicate that these currency market inefficiencies alone are probably not sufficient to explain the puzzle. The weakness of the domestic interest – exchange rate link also has important monetary policy implications, so that finding possible solutions to this puzzle should be placed high on the research agenda.

**Outline.** This paper proceeds as follows. Section 2 explains the country selection and provides some background information on their policy frameworks. In Section 3, a preliminary scatter plot analysis is presented, while Section 4 explains the estimates of the effects of domestic and foreign monetary policy shocks on the exchange rates using the baseline model. Section 5 shows evidence that very simple currency trading strategies actually deliver significant excess returns, which can be taken as an indication of market inefficiencies. A comprehensive classification of the policy events according to their inputted degree of exogeneity and the re-estimation of the specifications of Section 4 is shown in Section 6, where we analyze subsamples according to the degrees of exogeneity of the observations. As a robustness check, the estimations are repeated based on the variations against a weighted basket of global currencies. The robustness section also shows that domestic monetary policy shocks do not seem to lead to significant changes in expectations about the future rate of depreciation. Section 7 explores some conjectures about the causes of the surprising results, noting that many studies have taken output as given. It also discusses the related literature. The paper closes with some concluding remarks, indicating directions for further research.

## 2. Methodology

Our study of monetary policy events focuses on emerging economies that have an established floating exchange rate regime. By this we mean that the exchange rate regime has been classified as

<sup>4</sup> Mexico switched to a floating exchange rate regime in December 1994 and Brazil in January 1999. Chile did so in September 1999.

<sup>5</sup> As Zettelmeyer (2004) does, Kearns and Manners (2006) and Faust et al. (2007) also find the association of monetary contractions with currency appreciations in a selection of advanced economies.

<sup>6</sup> Note that the analysis in Eichenbaum and Evans (1995) and in Hnatkovska et al. (2011) relies on symmetry.

<sup>7</sup> As suggested by the theoretical contribution of Blanchard (2005) as well as the papers by Gonçalves and Guimarães (2011) and Hnatkovska et al. (2011), among others. Unfortunately, the latter two ignore exchange market interventions – which are prevalent in emerging economies – in their analysis.

floating by [Reinhart and Rogoff \(2004\)](#), and later updates of it by the IMF, for an uninterrupted period of at least 10 years. Emerging economies that fall into this category typically have a higher nominal exchange rate volatility than that of the United States and the Eurozone, even though most of them clearly practice a managed float. We further narrow the selection to countries for which we could obtain daily foreign exchange intervention data and in which the decision of the monetary policy committee meetings is announced after markets have already closed. These criteria left us with three emerging economies: Brazil, Mexico and Chile.

In the case of the *Banco Central do Brasil* and *Banco Central de Chile*, the policy announcement is made on the evening of the last day of the meeting and, in the case of *Banco de Mexico*, on the following morning, at 9:30 a.m., before the opening of markets.<sup>8</sup> All three countries that we study have also been operating within an inflation targeting framework for more than a decade (see [Gonçalves and Salles \(2008\)](#)) and use a short-term interest rate as their main policy instrument. We then proceeded to consider all pre-scheduled monetary policy committee meetings that took place in these countries between January 2003 and May 2011. This gave us a total of 238 policy events for which we could analyze the market developments on the day that followed the policy decision (79 in Brazil, 101 in Mexico and 58 in Chile).

To consider the surprise component of the decisions we examined how immediate changes in market interest rates relate to changes in the nominal exchange rates. Changes in market interest rates have been used as proxies for monetary surprises earlier, among others by [Kuttner \(2001\)](#), [Zettelmeyer \(2004\)](#), [Kearns and Manners \(2006\)](#) and [Faust et al. \(2007\)](#). The underlying idea is that, by focusing on changes in market rates, we filter the components that were already expected on the days prior to the announcements out.<sup>9</sup> As a robustness check, we also looked at the actual variation in the policy target and, in the case of Brazil, at the difference between the announced targets and previous expectations, as measured by market surveys.

### 3. Scatter plots

[Fig. 1](#) shows the scatter plots of exchange rate and market interest rate variations for each country. Throughout, the exchange rate is the domestic price of foreign currency. The plots on the left represent changes in the US Dollar exchange rate against changes in the 90 day interest rate swaps on the days that followed the monetary policy committee meetings. In the case of Mexico we used the 28 day interbank rate as a continuous daily series for the equivalent 90 day rate was not available.<sup>10</sup> The plots on the right are based on the actual changes in the targeted policy rate.<sup>11</sup> Throughout, we compute the exchange rate variations by comparing the last prices that are reported by *Bloomberg* for the day. The most striking feature of the plots is the large dispersion of the observations along the vertical dimension. The visual analysis suggests no clear relation between the exchange rates and market interest rate surprises in the case of Mexico and a possible positive relation in Brazil and in Chile – i.e. an association of increases in market interest rates with currency depreciations.

[Fig. 2](#) repeats the scatter plots separating the observations according to whether they occurred on a day in which the central bank intervened in the foreign exchange market or not. For this disaggregation we used actual intervention data, that was obtained from the respective Central Banks. It is interesting to note that, if anything, the relationship becomes even more positive in the sample without intervention. On the other hand, the intervention subsamples suggest a slightly negative relationship (as would be predicted by theory).

<sup>8</sup> In the case of the *Reserve Bank of South Africa*, in contrast, the announcement is made at 3:00 p.m. on the last day of the meetings, while markets are still open.

<sup>9</sup> The anticipated components of changes should not affect security prices, as prices already incorporated that piece of information.

<sup>10</sup> We do not use government bills as the issuance of these has not been continuous. In Brazil, DI x pré swap contracts are guaranteed by the futures exchange (*BMF&FBovespa*) and effectively replicate a zero-coupon bond.

<sup>11</sup> Note that in Mexico the 1 day interbank rate only became the main policy instrument in January 2008, which explains the smaller sample size in this case. Prior to that, the monetary policy committee used to announce a quantitative target (the *corto*).

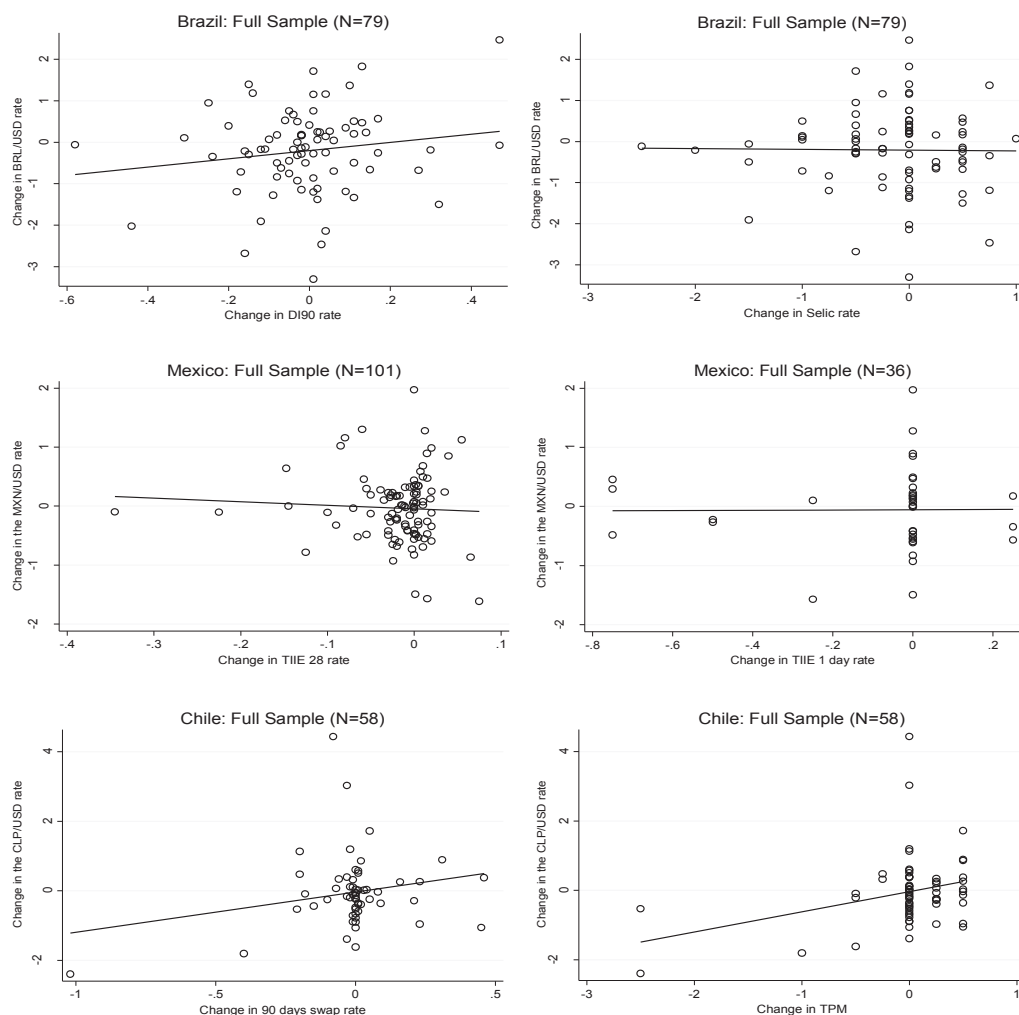


Fig. 1. Exchange rate and market interest rate variations.

## 4. Regression analysis

### 4.1. The effects of domestic monetary policy shocks

To assess the impact effect of monetary policy on the exchange rates we estimated

$$\Delta e_t = \alpha + \beta \Delta i_{n,t} + \gamma \Delta Z_t + \varepsilon_t \quad (1)$$

for each country, where  $\Delta e_t$  is the daily variation in the exchange rate,  $\Delta i_{n,t}$  represents the change in the  $n$ -month interest rate and  $\Delta Z_t$  stands for the change in eventual additional control variables on the day of the policy announcement – or the day after, if the policy announcement was made after market closure.  $\alpha$  captures the eventual trend depreciation over the sample period. The parameter of interest in this paper is  $\beta$ : if the usual textbook combination of uncovered interest parity and rational expectations were a good approximation of reality,  $\beta$  would be negative and statistically significant, meaning that a surprise monetary tightening would lead to an appreciation of the exchange rate.

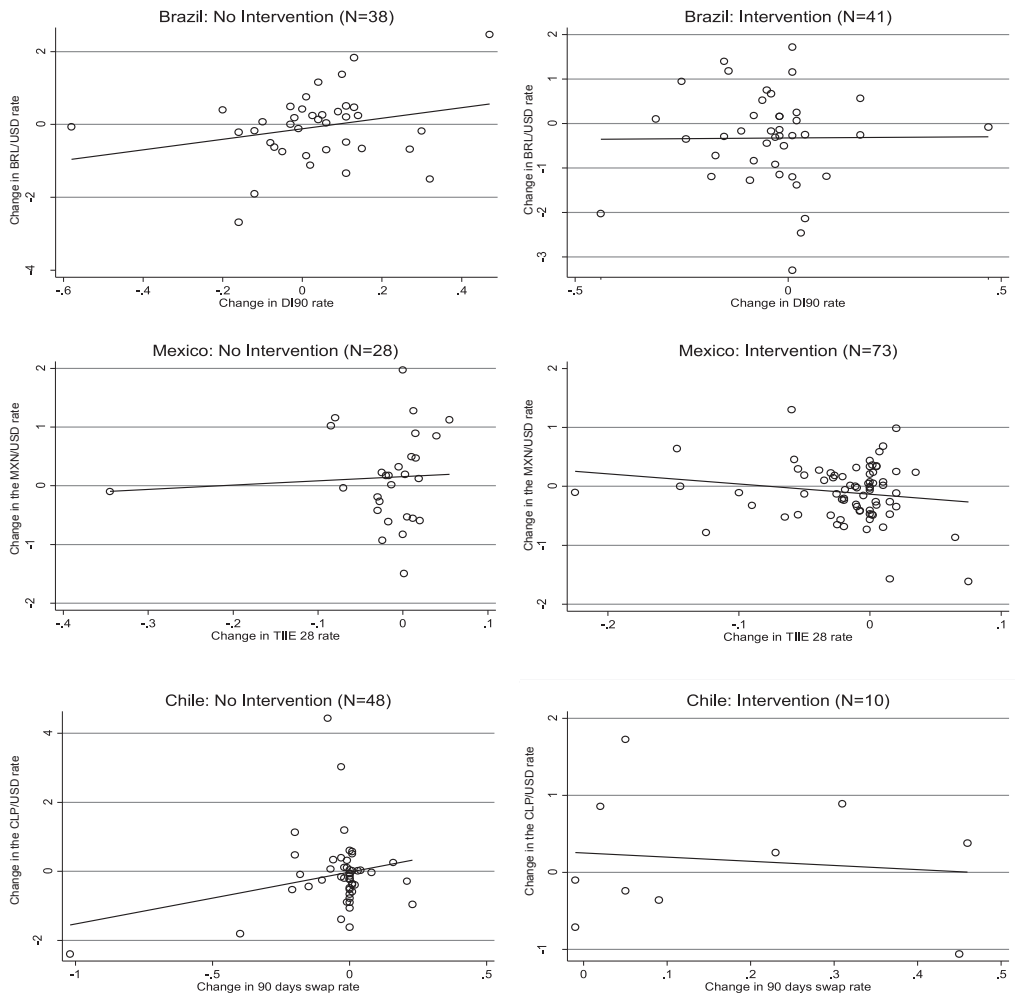


Fig. 2. The effects of exchange market intervention.

As already noted, the focus on the variation within a single day diminishes the potential that a problem of reverse causality may drive the estimation results. As we are focusing on the day following the policy announcement, the variation in the market interest rate should normally be dominated by the news regarding the monetary policy decision. Note that  $e_t$  and  $i_{n,t}$  are asset prices that should react to news instantaneously as long as markets are liquid. We use either a 1-month or a 3-month interest rate as the market for these instruments is sufficiently liquid and changes in short-term rates will reflect the surprise element in the new policy target. Obviously, the market interest rate will also reflect other developments that affected the money market within the same day, so that we cannot rule out the possibility of within-day endogeneity completely. This led us to re-estimate the above equation using the actual change in the policy target as an instrument. While the change in the targeted policy rate is unlikely to be correlated with non-policy factors that could have affected the 1 or the 3 month rate on the first opening day of markets, it is clearly correlated with the change in these rates.

><sup>12</sup> The Durbin and the Wu–Hausman test statistics clearly did not reject the null of exogeneity of the regressors in the cases of Brazil and Mexico for all specifications, backing the use of standard L.S. over I.V. This was not the case in Chile, where we report the results of the I.V. estimations instead.<sup>13</sup>

The first column in Table 1 shows the estimation output when no controls are included (as in the baseline case of Zettelmeyer), whereas the second column repeats the exercise including controls. The controls in  $Z_t$  intend to capture the changes in three factors: foreign monetary conditions, risk premia and international commodity prices. The change in the foreign monetary conditions is measured by the daily variation of the 3 month T-Bill rate – which is comparable to the 90 day swap rates in the cases of Brazil and Chile,<sup>14</sup> whereas in the case of Mexico we use the change in the 1 month US Dollar LIBOR rate to compare with the change in the 28 day interbank rate. The change in the country specific risk premium is proxied by the daily change in the VIX rate, as the strong correlation of the VIX with changes in sovereign credit default spreads has been documented earlier (see Pan and Singleton (2008)).<sup>15</sup> Indeed, the change in the VIX rate turns out to be highly correlated with the daily change in the 5 year CDS spreads also within our sample period. (The correlation over the sample period was 0.584 for Brazil, 0.702 for Mexico and 0.708 for Chile.) Using the CDS spread directly to capture the risk premium would have the potential of compounding the problem of reverse causality – which previous authors have pointed out as being the culprit for the association of interest rate hikes with depreciations in studies that are based on monthly or quarterly data. While in principle the exchange rate variation could have a direct effect on the CDS spread within the same day, it is highly unlikely that the exchange rates of any of these countries drive the VIX spread. In any case, as a robustness check, we re-ran the regressions using the CDS spreads instead (see Section 6.5). Finally, the change in international commodity prices was measured by the daily variation in the CRB index, a commodity price index that is calculated on a daily basis by the Commodity Research Bureau, in the cases of Brazil and Chile. As in the case of Mexico oil is the only commodity that lists among the top 20 export products of the country, we used the daily West Texas Intermediate crude oil price variation instead of an index.<sup>16</sup>

The third and the fourth columns in Table 1 repeat the estimations restricting the subsample to the days in which the respective Central Banks did not intervene in the foreign exchange market. Throughout, our control variables do attain the expected sign wherever they turn out to be statistically significant. Increases in international interest rates, in the VIX rate and decreases in international commodity prices lead to depreciations of the currencies. The reduction in risk appetite, proxied by the increase in the VIX rate, leads to statistically significant depreciations in all three currencies, most notably so in Brazil. Finally, increases in commodity prices lead to very significant appreciations of the Chilean Peso, as one would expect. The much better fit of the regressions led us to adopt the version with the control variables as the baseline specification for the remainder of this paper.

As for our main variable of interest,  $\beta$ , we never manage to attain anything close to the negative sign predicted by theory in a way that is significant. On the contrary, the point estimates indicate that an unexpected monetary tightening that leads to a 100 b.p. increase in the 90 day DI swap rate would lead to a 1.0–1.6% depreciation of the Brazilian Real on impact. In the case of Mexico, a hike that causes a 100 b.p. increase in the 28 day TIIE rate, would lead to smaller variations in the value of the Peso. Based on the robust standard errors, neither effect differs significantly from zero. The estimates for Chile are the most surprising ones: our point estimates suggest that a tightening of 100 b.p. in the *tasa de política monetaria* would lead to a 2.2–2.4% depreciation of the Chilean Peso. This effect turns out to be significant at the 1% confidence level in all four cases.<sup>17</sup> Indeed, this finding suggests that the positive

<sup>12</sup> For our sample of policy events, the correlation is 0.395 for Brazil, 0.621 for Mexico and 0.761 for Chile.

<sup>13</sup> Note that, since the models are not overidentified, the Sargan–Hansen test does not apply.

<sup>14</sup> In the Chilean case the daily series starts in May 2006, which is why we focus on the policy events that took place after this date.

<sup>15</sup> The VIX index is a measure of equity market volatility that is computed by the Chicago Board Options Exchange.

<sup>16</sup> In Brazil, 15 of the 20 top export products are commodities, whereas in Chile all of them are commodities.

<sup>17</sup>  $\beta$ s are also positive and significant under OLS, when the variation in the 90 day swap rate is used. The Durbin and the Wu–Hausman tests however reject the null hypothesis that regressors are exogenous in this specification.

**Table 1**

The impact of monetary policy on the exchange rate.

|                              | Full sample     |                 | No intervention |                 |
|------------------------------|-----------------|-----------------|-----------------|-----------------|
|                              | L.S.            | L.S.            | L.S.            | L.S.            |
| D.V.: change in BRL/USD rate |                 |                 |                 |                 |
| Brazil                       |                 |                 |                 |                 |
| <b>d (DI 90)</b>             | <b>0.989</b>    | <b>1.368</b>    | <b>1.465</b>    | <b>1.599</b>    |
|                              | <b>0.827</b>    | <b>0.825</b>    | <b>1.234</b>    | <b>1.186</b>    |
| d (3 month T Bill)           |                 | 0.672           |                 | 3.764           |
|                              |                 | 2.733           |                 | 3.297           |
| d (VIX)                      |                 | 0.334***        |                 | 0.298***        |
|                              |                 | 0.088           |                 | 0.109           |
| d (price of commodities)     |                 | 0.068           |                 | 0.365           |
|                              |                 | 0.232           |                 | 0.255           |
| No. of observations          | 79              | 79              | 38              | 38              |
| R2                           | 0.025           | 0.210           | 0.069           | 0.293           |
| RMSE                         | 1.003           | 0.920           | 0.930           | 0.846           |
| F/chi2                       | 1.43            | 4.42***         | 1.41            | 3.66**          |
| Durbin score (chi 2)         | 0.833           | 0.557           | 0.008           | 0.093           |
| Wu-Hausman (F)               | 0.772           | 0.497           | 0.008           | 0.082           |
| D.V.: change in MXN/USD rate |                 |                 |                 |                 |
| Mexico                       |                 |                 |                 |                 |
| <b>d (TIE 28)</b>            | <b>-0.609</b>   | <b>-0.439</b>   | <b>0.730</b>    | <b>1.192</b>    |
|                              | <b>1.034</b>    | <b>1.086</b>    | <b>1.114</b>    | <b>1.464</b>    |
| d (1 m USD LIBOR)            |                 | 1.711*          |                 | 1.603*          |
|                              |                 | 0.874           |                 | 0.855           |
| d (VIX)                      |                 | 0.163***        |                 | 0.164*          |
|                              |                 | 0.053           |                 | 0.093           |
| d (price of WTI crude oil)   |                 | -0.030          |                 | -0.032          |
|                              |                 | 0.018           |                 | 0.057           |
| No. of observations          | 101             | 101             | 28 (small N)    | 28 (small N)    |
| R2                           | 0.003           | 0.240           | 0.005           | 0.336           |
| RMSE                         | 0.587           | 0.521           | 0.790           | 0.686           |
| F/chi2                       | 0.35            | 3.39**          | 0.43            | 1.55            |
| Durbin score (chi 2)         | 0.083           | 0.421           | small N         | small N         |
| Wu-Hausman (F)               | 0.076           | 0.354           | small N         | small N         |
| D.V.: change in CLP/USD rate |                 |                 |                 |                 |
|                              | Full sample     |                 | No intervention |                 |
|                              | L.V.            | L.V.            | L.V.            | L.V.            |
| Chile                        |                 |                 |                 |                 |
| <b>d (swap 90)</b>           | <b>2.258***</b> | <b>2.206***</b> | <b>2.428***</b> | <b>2.296***</b> |
|                              | <b>0.445</b>    | <b>0.439</b>    | <b>0.428</b>    | <b>0.443</b>    |
| d (3 month T Bill)           |                 | -2.592          |                 | -3.999          |
|                              |                 | 2.591           |                 | 2.797           |
| d (VIX)                      |                 | 0.200**         |                 | 0.185*          |
|                              |                 | 0.089           |                 | 0.096           |
| d (price of commodities)     |                 | -0.493***       |                 | -0.377**        |
|                              |                 | 0.170           |                 | 0.170           |
| No. of observations          | 58              | 58              | 48              | 48              |
| R2                           | 0.004           | 0.434           | 0.038           | 0.469           |
| RMSE                         | 1.034           | 0.764           | 1.031           | 0.766           |
| F/chi2                       | 25.79***        | 49.33***        | 32.17***        | 46.22***        |
| Durbin score (chi 2)         | 7.961***        | 7.184***        | 5.235**         | 5.871**         |
| Wu-Hausman (F)               | 3.138*          | 4.707**         | 2.252           | 3.619*          |

Note: Robust standard errors. \*, \*\* and \*\*\* denote statistical significance at the 10%, 5% and 1% confidence levels, respectively.

slope that was apparent in the scatter plots is not accidental in the case of the Chilean Peso.<sup>18</sup> In any case, the estimated interest rate coefficient is always positive in the sample that excludes government interventions, for all three countries. This results turns out to hold for all specifications that we have used in this paper.

<sup>18</sup> Given the low pass-through from the exchange rate to inflation, this effect however is likely to have only minor implications for inflation.

The Chilean case is very telling because it would seem to invalidate attributing the failure to find an association of interest rate hikes with currency appreciations in these emerging markets to a problem of fiscal dominance. Chile has had the highest possible short-term sovereign credit rating at Standard & Poor's since 1995 and its debt/GDP ratio of 8% is one of the lowest in the world. Fiscal dominance does not explain the pattern of the results in Brazil or Mexico either. It is important to note that the average 1 year CDS spread between 2003 and 2011 was of 123 bp in the case of Brazil, 58 bp in the case of Mexico and 32 bp in the case of Chile (the relevant 3-month or 1-month rate would presumably be even smaller, if available). Fig. 3 shows the variation in the 5 year CDS spreads against interest variations for each of these three countries. It is apparent that – contrary to what a situation of fiscal dominance would suggest – there is no significant relation between the variables.

Summing up, we fail to find the link of interest hikes with currency appreciations or any evidence that would square, for instance, with the prediction of standard rational expectations models that are based on the UIP when looking at impact effects. In particular, note that, if anything, dropping all events in which the monetary authorities intervened in the foreign exchange market on the day that followed the policy meeting moves us even further away from the negative coefficient that one would have expected a priori.

#### 4.2. The effects of U.S. monetary policy shocks

In an attempt to come closer to an understanding of the source of the puzzling results that are reported in Table 1, we analyzed the response of exchange rates to changes in the Fed Funds rate following FOMC meeting days as well.

Table 2 reports the estimated sensitivities of the same exchange rates to changes in the Fed Funds rate on FOMC meeting days. There were a total of 67 FOMC meetings during our sample period. Obviously, to the extent that US interest rate changes may induce partially offsetting changes in the domestic rates, responses may be dampened. What becomes clear, however, is that whenever responses are significant, they do attain the expected sign, i.e., the Brazilian Real and the Mexican Peso do depreciate in response to an unexpected increase in the Fed Funds rate. The asymmetry relative to the response to domestic rate changes suggests that the responses we found in the previous subsection are more likely to be related to domestic factors.

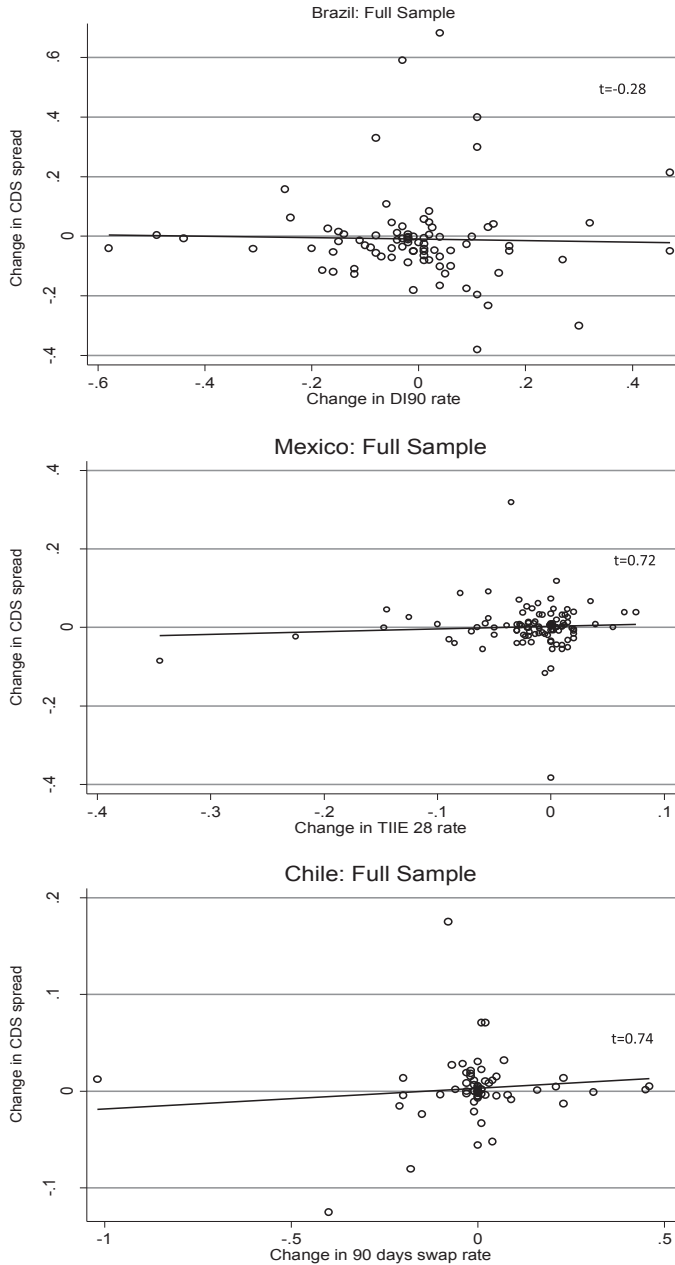
There is no indication, however, that the response pattern is related to any anomalous behavior of the yield curve. Even though liquidity at the longer end of the curve is significantly lower than that at the shorter end in developing countries, the estimates that are reported in Table 3 show that increases in the shorter market interest rates on policy announcement days are accompanied by increases at the longer end of the curve. As one would expect, the effect is stronger at the 1 year horizon than at the 2 year horizon.

### 5. Testing for foreign exchange market efficiency

The reaction of exchange rates to monetary policy shocks (or absence of it), seems to indicate that there may be departures from foreign exchange market efficiency at play. An interesting study by Olmo and Pilbeam (2011), however, makes the case that UIP holding is actually not a necessary condition for foreign exchange market efficiency. Instead, they argue that one should test the efficient market hypothesis by checking whether simple trading strategies that involve switching between currencies are actually able to generate excess returns that are significant. They point out that, in some cases, a set of well-known trading strategies is unable to deliver profits that are significant, in spite of a clear rejection of UIP. Simple trading strategies however did deliver statistically significant profits in the Sterling-Dollar and the Yen-Dollar markets in their sample.

While our findings have much broader implications than a simple rejection of the UIP – in that we actually reject the simple association of surprise rate hikes with appreciations more generally – we were able to test for foreign exchange market efficiency in Brazil and Mexico using the profitability tests of the very same strategies that Olmo and Pilbeam use in their study.<sup>19</sup> More

<sup>19</sup> We were not able to find an appropriate 1 month local instrument that would allow us to perform the same profitability based test for the Chilean market for a reasonably long sample period.



**Fig. 3.** Risk premia and interest rate variations.

specifically, we tested first whether a passive strategy that receives the 1 month interest rate in Brazil or Mexico is able to generate relevant excess returns (measured by the estimated parameter  $\alpha$ ) relative to receiving the 1 month US Dollar rate. The second strategy that we tested, was the momentum based strategy that simply switches funds to the market that delivered the largest return in the previous month (measured in US Dollars). Finally, we also checked the returns of a carry trade strategy, by comparing the payoffs from always investing in the high interest rate

**Table 2**

The impact of U.S. monetary policy on the exchange rate.

| D.V.: change in BRL/USD rate        | Full sample     |                | No intervention |                 |
|-------------------------------------|-----------------|----------------|-----------------|-----------------|
|                                     | I.V.            | I.V.           | I.V.            | I.V.            |
| <b>Brazil</b>                       |                 |                |                 |                 |
| <b>d (3 month T-Bill)</b>           | <b>0.481***</b> | <b>0.314**</b> | <b>0.337</b>    | <b>0.540*</b>   |
|                                     | <b>0.162</b>    | <b>0.127</b>   | <b>0.259</b>    | <b>0.301</b>    |
| d (DI 90)                           |                 | 0.007          |                 | 0.005           |
|                                     |                 | 0.011          |                 | 0.011           |
| d (VIX)                             |                 | 0.248***       |                 | 0.287***        |
|                                     |                 | 0.062          |                 | 0.094           |
| d (price of commodities)            |                 | 0.018          |                 | 0.607           |
|                                     |                 | 0.231          |                 | 0.483           |
| No. of observations                 | 67              | 67             | 25              | 25              |
| R2                                  | 0.076           | 0.233          | 0.004           | 0.184           |
| RMSE                                | 1.032           | 0.888          | 1.150           | 1.111           |
| F/chi2                              | 8.78***         | 23.88***       | 1.69            | 14.44***        |
| Durbin score (chi 2)                | 3.105*          | 1.696          | 1.472           | 2.092           |
| Wu–Hausman (F)                      | 5.828**         | 1.903          | 2.119           | 1.385           |
| <b>D.V.: Change in MXN/USD rate</b> |                 |                |                 |                 |
| <b>Mexico</b>                       |                 |                |                 |                 |
| <b>d (3 month T-Bill)</b>           | <b>0.248**</b>  | <b>0.125</b>   | <b>0.512***</b> | <b>0.403***</b> |
|                                     | <b>0.108</b>    | <b>0.098</b>   | <b>0.192</b>    | <b>0.091</b>    |
| d (TIIE)                            |                 | 0.005          |                 | –0.032          |
|                                     |                 | 0.011          |                 | 0.035           |
| d (VIX)                             |                 | 0.155***       |                 | 0.191***        |
|                                     |                 | 0.050          |                 | 0.035           |
| d (price of WTI crude oil)          |                 | –0.036         |                 | 0.028           |
|                                     |                 | 0.038          |                 | 0.068           |
| No. of observations                 | 67              | 67             | 25              | 25              |
| R2                                  | 0.040           | 0.343          | 0.176           | 0.205           |
| RMSE                                | 0.537           | 0.444          | 0.863           | 0.658           |
| F/chi2                              | 5.28**          | 19.64***       | 7.12***         | 60.04***        |
| Durbin score (chi 2)                | 1.395           | 0.231          | 2.737*          | 2.315           |
| Wu–Hausman (F)                      | 2.313           | 0.444          | 23.235***       | 4.511**         |
| <b>D.V.: Change in CLP/USD rate</b> |                 |                |                 |                 |
| <b>Chile</b>                        |                 |                |                 |                 |
| <b>d (3 month T-Bill)</b>           | <b>0.047</b>    | <b>–0.033</b>  | <b>0.070</b>    | <b>–0.080</b>   |
|                                     | <b>0.063</b>    | <b>0.096</b>   | <b>0.052</b>    | <b>0.113</b>    |
| d (swap 90)                         |                 | 0.021          |                 | 0.035**         |
|                                     |                 | 0.017          |                 | 0.017           |
| d (VIX)                             |                 | 0.024          |                 | 0.041           |
|                                     |                 | 0.031          |                 | 0.036           |
| d (price of commodities)            |                 | –0.593***      |                 | –0.677**        |
|                                     |                 | 0.216          |                 | 0.284           |
| No. of observations                 | 67              | 40             | 60              | 33              |
| R2                                  | 0.046           | 0.187          | 0.033           | 0.258           |
| RMSE                                | 0.640           | 0.684          | 0.619           | 0.634           |
| F/chi2                              | 0.56            | 14.74***       | 1.79            | 14.11***        |
| Durbin score (chi 2)                | 3.500*          | 2.729*         | 1.066           | 0.566           |
| Wu–Hausman (F)                      | 5.194**         | 3.331*         | 1.019           | 0.495           |

Note: Robust standard errors. \*, \*\* and \*\*\* denote statistical significance at the 10%, 5% and 1% confidence levels, respectively.

currency with the returns from opting for the low interest rate currency. In all cases, returns were computed taking the withholding taxes of each country into account.<sup>20</sup>

We use monthly returns for a sample period that starts in January 1995 and extends to October 2013.<sup>21</sup> The results for each combination of strategy and currency pair are shown in Table 4. In the case

<sup>20</sup> The fourth strategy suggested by Olmo and Pilbeam – which is always to be positioned in the currency that is being traded at a forward discount – would be equivalent to always investing in the US Dollar, as both, the Real and the Peso were at a forward premium on the last day of every month.

<sup>21</sup> In the case of Mexico, because of data availability, the sample period starts in July 2004.

**Table 3**

The impact of monetary policy on long rates.

| D.V.: Change in the long rate | 360 days swap   |                 |               | 720 days swap   |              |              |
|-------------------------------|-----------------|-----------------|---------------|-----------------|--------------|--------------|
|                               | Brazil          | Mexico          | Chile         | Brazil          | Mexico       | Chile        |
| <b>d (short swap rate)</b>    | <b>0.940***</b> | <b>0.465***</b> | <b>0.360*</b> | <b>0.791***</b> | <b>0.201</b> | <b>0.228</b> |
|                               | <b>0.163</b>    | <b>0.173</b>    | <b>0.205</b>  | <b>0.217</b>    | <b>0.151</b> | <b>0.148</b> |
| Constant                      | 0.027*          | 0.011           | −0.011        | 0.036*          | −0.006       | −0.013       |
|                               | 0.015           | 0.010           | 0.013         | 0.020           | 0.011        | 0.011        |
| No. of observations           | 78              | 94              | 56            | 77              | 86           | 56           |
| R2                            | 0.568           | 0.249           | 0.317         | 0.381           | 0.062        | 0.207        |
| RMSE                          | 0.133           | 0.099           | 0.104         | 0.164           | 0.100        | 0.087        |
| F/chi2                        | 33.31***        | 7.27***         | 3.09*         | 13.30***        | 1.77         | 2.37         |

Note: Robust standard errors. \*, \*\* and \*\*\* denote statistical significance at the 10%, 5% and 1% confidence levels, respectively.

of Brazil, we also report the results of the tests for a subsample that ends in February 2008 – before the first introduction of the tax on foreign portfolio inflows (the IOF). While the tax did not apply to money that was already in local markets, and there is an ongoing controversy on whether the tax (which was scrapped in June 2013) was effective, this wedge would tend to limit arbitrage activity between currency markets as it imposed a significant cost on new entrants.

What is clear from the table is that each simple trading strategy would have produced economically and statistically significant excess returns in the case of Brazil. Yearly excess returns in USD vary between 7.19% and 11.75%, depending on the strategy. In the case of the passive strategy and the carry trade, the tests indicate that there is no evidence of serial correlation or heteroskedasticity in the residuals of the return differentials. The Breusch–Godfrey LM test however points to autocorrelation in residuals arising from the momentum strategy returns. In any case, the Table shows that the excess returns are also clearly significant when the heteroskedasticity and autocorrelation consistent covariance matrix is used.

In the case of Mexico, the estimated  $\alpha$  is always positive as well, indicating excess returns of the 3 different strategies. The returns however are only significant from a statistical viewpoint in the case of the momentum strategy (which is also the strategy for which the residuals are well behaved).

**Table 4**

Profitability based tests of market efficiency.

| Currency pair            | Strategy         | $\alpha$             | Obs | Residuals  |         |        |
|--------------------------|------------------|----------------------|-----|------------|---------|--------|
|                          |                  |                      |     | JB         | BG-LM   | ARCH   |
| Brazilian Real—US Dollar | Passive          | 0.0079**<br>(0.014)  | 226 | 1787.7***  | 0.02    | 0.82   |
| Brazilian Real—US Dollar | Momentum         | 0.0058**<br>(0.019)  | 225 | 18972.4*** | 6.34*** | 0.05   |
| Brazilian Real—US Dollar | Carry            | 0.0078**<br>(0.016)  | 225 | 1763.5***  | 0.02    | 0.81   |
| Brazilian Real—US Dollar | w/o IOF Passive  | 0.0093***<br>(0.008) | 158 | 2480.3***  | 0.69    | 0.26   |
| Brazilian Real—US Dollar | w/o IOF Momentum | 0.0064**<br>(0.041)  | 157 | 11926.1*** | 5.87*** | 0.06   |
| Brazilian Real—US Dollar | w/o IOF Carry    | 0.0092***<br>(0.009) | 157 | 2433.5***  | 0.68    | 0.26   |
| Mexican Peso—US Dollar   | Passive          | 0.0028<br>(0.368)    | 112 | 127.5***   | 0.49    | 4.76** |
| Mexican Peso—US Dollar   | Momentum         | 0.0032**<br>(0.042)  | 111 | 27.4***    | 0.89    | 0.44   |
| Mexican Peso—US Dollar   | Carry            | 0.0028<br>(0.384)    | 111 | 122.9***   | 0.49    | 4.65** |

Note: Returns were computed taking withholding taxes in each country into account. p values based on Newey–West std. errors are reported in parenthesis. \*, \*\* and \*\*\* denote statistical significance at the 10%, 5% and 1% confidence levels, respectively.

All in all, the results of these profitability based tests indicate that there may be important departures from market efficiency in these emerging foreign exchange markets. These findings are in line with those that emerged in the literature that followed [Burnside et al. \(2006\)](#), who have shown significant returns for carry trade strategies even after controlling for risk.

While we clearly do tend to find evidence of exchange rate market inefficiencies, the results of the previous section pointed to an important asymmetry. The fact that the exchange rate does react to changes in US monetary policy in the way that conventional models suggest in the cases of Brazil and Mexico – whereas there is an apparent lack of sensitivity of exchange rates to domestic interest rate changes – suggests that currency market inefficiencies on their own are probably not sufficient to explain the complete pattern of responses that we find.<sup>22</sup>

## 6. Robustness

To assess the robustness of the findings of the previous sections, we proceed to analyze first whether the results persist once we drop “contaminated events” – following the narrative approach of [Zettelmeyer \(2004\)](#). Second, we check whether results change if effective exchange rate variations are used instead of bilateral variations, where effective exchange rates measure the domestic price of a weighted basket of the four main global currencies. Third, we test whether a non-linear specification of the relation between the interest rate and the exchange rate would be able to provide support for the usual association of surprise rate hikes with currency appreciations. We then proceed to show that our results do not seem to be driven by our choice of *proxy* for monetary innovations. Fifth, we find that estimates for our variable of interest  $\beta$  are qualitatively similar when we use country specific CDS spreads as indicators of risk (even though the risk of reverse causality is increased under this specification). Sixth, we show that there is no indication of response asymmetry between positive and negative rate surprises – which constitutes further evidence against a fiscal explanation. Finally, we document that domestic interest rate shocks also do not affect expectations of exchange rate depreciation in a significant way.

### 6.1. Exogeneity classes

Zettelmeyer observed that a researcher may fail to find evidence in favor of the UIP condition because of data contamination. While this issue should be much more serious in studies at lower frequencies, in our setting this could occur because market interest rates may react to other changes, besides the changes in the policy rate, within the day that follows the policy announcement. For this reason, we scrutinized the news that surrounded all pre-scheduled monetary policy committee meetings of this study. We did this by analyzing all country news that were made public by *Bloomberg* between the time of the policy announcement and the following market closure. The cover pages of major newspapers on the days that followed the monetary policy committee meetings were also analyzed. In the case of Brazil we used the archives of *Folha de São Paulo*, while in the case of Mexico the archives of *El Universal* were used. Both newspapers typically give ample coverage to monetary policy decisions. For the case of Chile, we used *Bloomberg* additionally to search for major global events that may have affected the local money market on the days that followed the monetary policy decisions.

The overall strategy was to highlight any data release that was related either to GDP, inflation or unemployment, as well as reports concerning the state of the global economy and other measures that clearly could have had an economic impact that appeared in the media outlets mentioned above. We identified that 21.8% of the policy events could suffer from potential contamination in the case of Brazil, 22.7% in the case of Mexico and 20.7% in the case of Chile. Each of the policy events was then classified according to whether there were other news that may have affected the money market on that particular date. We considered that the release of GDP and CPI data represented the highest degree of contamination, because of their importance within an inflation targeting framework, and classified the events that coincided with such releases as being of the exogeneity class C. Changes to the minimum

<sup>22</sup> Note that the comparison of [Tables 1](#) and [2](#) shows that there is also response asymmetry in the case of Chile.

**Table 5**

BCB monetary policy meetings.

| Meeting    | Exogeneity class | Intervention | Event                                           | $\Delta I90$<br>(b.p.) | $\Delta e$ (BRL/USD) |
|------------|------------------|--------------|-------------------------------------------------|------------------------|----------------------|
| 1/22/2003  | A                | 0            |                                                 | 30                     | −0.184               |
| 2/19/2003  | A                | 0            |                                                 | −10                    | 0.069                |
| 3/19/2003  | A                | 0            |                                                 | 11                     | 0.202                |
| 4/23/2003  | A                | 0            |                                                 | 9                      | 0.350                |
| 5/21/2003  | A                | 0            |                                                 | 6                      | −0.700               |
| 6/18/2003  | A                | 0            |                                                 | −3                     | 0.000                |
| 7/23/2003  | A                | 0            |                                                 | −8                     | −0.500               |
| 8/20/2003  | A                | 0            |                                                 | −1                     | −0.117               |
| 9/17/2003  | A                | 0            |                                                 | −16                    | −0.213               |
| 10/22/2003 | A                | 0            |                                                 | 6                      | 0.042                |
| 11/19/2003 | A                | 0            |                                                 | −58                    | −0.061               |
| 12/17/2003 | A                | 0            |                                                 | 4                      | 0.136                |
| 1/21/2004  | A                | 1            |                                                 | 47                     | −0.077               |
| 2/18/2004  | A                | 0            |                                                 | 11                     | 0.509                |
| 3/17/2004  | A                | 0            |                                                 | −12                    | −0.172               |
| 4/14/2004  | A                | 0            |                                                 | 4                      | 1.157                |
| 5/19/2004  | A                | 0            |                                                 | 47                     | 2.466                |
| 6/16/2004  | C                | 0            | Government defeat in Senate raises minimum wage | 5                      | 0.265                |
| 7/21/2004  | B                | 0            | Unemployment statistics released                | 0                      | 0.419                |
| 8/18/2004  | A                | 0            |                                                 | −2                     | 0.185                |
| 9/15/2004  | A                | 0            |                                                 | −7                     | −0.620               |
| 10/20/2004 | A                | 0            |                                                 | 27                     | −0.678               |
| 11/17/2004 | A                | 0            |                                                 | 13                     | 0.471                |
| 12/15/2004 | A                | 1            |                                                 | 17                     | 0.569                |
| 1/19/2005  | A                | 0            |                                                 | 14                     | 0.240                |
| 2/16/2005  | A                | 1            |                                                 | −5                     | −0.446               |
| 3/16/2005  | A                | 0            |                                                 | 32                     | −1.502               |
| 4/20/2005  | A                | 0            |                                                 | 11                     | −0.490               |
| 5/18/2005  | A                | 0            |                                                 | 15                     | −0.659               |
| 6/15/2005  | A                | 0            |                                                 | −5                     | −0.751               |
| 7/20/2005  | A                | 0            |                                                 | 1                      | 0.752                |
| 8/17/2005  | C                | 0            | Government overturns minimum wage               | 13                     | 1.831                |
| 9/14/2005  | A                | 0            |                                                 | 1                      | −0.862               |
| 10/19/2005 | A                | 1            |                                                 | −25                    | 0.947                |
| 11/23/2005 | B                | 1            | Unemployment statistics released                | 2                      | 0.067                |
| 12/14/2005 | A                | 1            |                                                 | 1                      | 1.718                |
| 1/18/2006  | A                | 1            |                                                 | −18                    | −1.196               |
| 3/8/2006   | A                | 1            |                                                 | −8                     | −0.835               |
| 4/19/2006  | A                | 1            |                                                 | 2                      | 0.251                |
| 5/31/2006  | A                | 0            |                                                 | −16                    | −2.683               |
| 7/19/2006  | A                | 1            |                                                 | −4                     | 0.671                |
| 8/30/2006  | C                | 0            | GDP statistics released                         | −20                    | 0.398                |
| 10/18/2006 | A                | 1            |                                                 | −8                     | 0.178                |
| 11/29/2006 | C                | 1            | GDP statistics released                         | −11                    | −0.171               |
| 1/24/2007  | A                | 0            |                                                 | 0                      | 0.244                |
| 3/7/2007   | A                | 1            |                                                 | 1                      | −0.275               |
| 4/18/2007  | A                | 1            |                                                 | −4                     | −0.172               |
| 6/6/2007   | A                | 1            |                                                 | −15                    | −0.295               |
| 7/18/2007  | A                | 1            |                                                 | −2                     | −0.280               |
| 9/5/2007   | A                | 0            |                                                 | 2                      | −1.118               |
| 10/17/2007 | A                | 1            |                                                 | −44                    | −2.026               |
| 12/5/2007  | A                | 1            |                                                 | 1                      | −1.202               |
| 1/23/2008  | B                | 1            | Unemployment statistics released                | 4                      | −2.138               |
| 3/5/2008   | A                | 1            |                                                 | −6                     | 0.527                |
| 4/16/2008  | A                | 1            |                                                 | 17                     | −0.253               |
| 6/4/2008   | A                | 1            |                                                 | −2                     | −0.141               |

Table 5 (continued)

| Meeting    | Exogeneity class | Intervention | Event                                                        | $\Delta DI90$<br>(b.p.) | $\Delta e$ (BRL/USD) |
|------------|------------------|--------------|--------------------------------------------------------------|-------------------------|----------------------|
| 7/23/2008  | A                | 1            |                                                              | –24                     | –0.347               |
| 9/10/2008  | C                | 0            | GDP statistics released & Lehman Brothers incident           | 10                      | 1.370                |
| 10/29/2008 | C                | 0            | Fed Funds rate slashed to 1% & \$ 30 bn credit line from Fed | 11                      | –1.336               |
| 12/10/2008 | B                | 1            | Income tax reduction                                         | 1                       | –3.298               |
| 1/21/2009  | A                | 1            |                                                              | –17                     | –0.719               |
| 3/11/2009  | A                | 0            |                                                              | –12                     | –1.902               |
| 4/29/2009  | A                | 0            |                                                              | –3                      | 0.496                |
| 6/10/2009  | A                | 1            |                                                              | –31                     | 0.104                |
| 7/22/2009  | B                | 1            | Unemployment statistics released                             | 4                       | –0.247               |
| 9/2/2009   | A                | 1            |                                                              | 2                       | –1.383               |
| 10/21/2009 | B                | 1            | Unemployment statistics released                             | –3                      | –0.922               |
| 12/9/2009  | C                | 1            | GDP statistics released                                      | –3                      | –0.312               |
| 1/27/2010  | A                | 1            |                                                              | –5                      | 0.753                |
| 3/17/2010  | A                | 1            |                                                              | –15                     | 1.397                |
| 4/28/2010  | A                | 1            |                                                              | 9                       | –1.190               |
| 6/9/2010   | A                | 1            |                                                              | 3                       | –2.465               |
| 7/21/2010  | B                | 1            | Unemployment statistics released                             | –9                      | –1.275               |
| 9/1/2010   | A                | 1            |                                                              | –2                      | –1.145               |
| 10/20/2010 | B                | 1            | Unemployment statistics released                             | 1                       | 1.156                |
| 12/8/2010  | C                | 1            | GDP statistics released                                      | –14                     | 1.183                |
| 1/19/2011  | A                | 1            |                                                              | –2                      | 0.162                |
| 3/2/2011   | C                | 1            | GDP statistics released                                      | –1                      | –0.501               |
| 4/20/2011  | A                | 1            |                                                              | –2                      | 0.160                |

wage – which clearly have strong fiscal implications in the case of Brazil – as well as events in the international economy that made it to the front page of the major national newspapers, as for instance the *Lehman Brothers* incident or the slashing of the Fed Funds rate in October 2008 were also put into this class. We considered that the publication of unemployment statistics represented a lower degree of contamination, assigning it an exogeneity classification B, due to the lower weight that this variable has in economies where only a part of the labour market is formal. The remaining policy events, which had no overlap with other significant economic news, were assigned to exogeneity class A. The complete list of the meetings and their classification can be seen in Tables 5–7.

Table 8 lists the regression outputs for the cases where contaminated observations are dropped from the sample. We report the results for the cases in which only observations of the exogeneity class C were dropped as well as results for when we dropped both, classes B and C. By and large, it is apparent that the results repeat those reported in Table 1 quite closely. In particular, in the case of Chile, the association of interest rate hikes with depreciations of the Peso remains significant at the 1% confidence level throughout and the point estimates remain largely unaffected.<sup>23</sup> On the whole, we do not find an association between interest rate hikes and appreciations even if we restrict our sample to contain only the observations of exogeneity class A. Indeed, we did not find such association even when we additionally dropped the observations that were associated with market volatility, i.e. the observations in which either the VIX rate, the CDS spread or the CRB index varied by more than a percentage point in absolute terms.

## 6.2. Effective exchange rates

Another possibility is that the above failure to find evidence in favor of the conventional view is that so far we have looked exclusively at the bilateral exchange rates against the US Dollar. To investigate whether this is the case, we checked whether results change when effective exchange rates are used as dependent variables instead. For this purpose, the daily variations of the national currencies against

<sup>23</sup> Note that the two events with the highest variations in the CLP/USD rate were classified as C, as they were associated with the announcements of intervention programs.

**Table 6**

Banxico monetary policy meetings.

| Meeting    | Exogeneity class | Intervention | Event                                               | $\Delta \text{TIIE28}$ (b.p.) | $\Delta e$ (MXN/USD) |
|------------|------------------|--------------|-----------------------------------------------------|-------------------------------|----------------------|
| 1/10/2003  | A                | 0            |                                                     | 6                             | 1.127                |
| 1/24/2003  | A                | 0            |                                                     | −8                            | 1.158                |
| 2/7/2003   | C                | 0            | CPI statistics released                             | −9                            | 1.023                |
| 2/21/2003  | C                | 0            | GDP statistics released                             | −3                            | −0.193               |
| 3/28/2003  | A                | 0            |                                                     | −7                            | −0.033               |
| 4/25/2003  | A                | 0            |                                                     | −35                           | −0.098               |
| 5/23/2003  | A                | 1            |                                                     | −23                           | −0.102               |
| 6/27/2003  | A                | 1            |                                                     | −13                           | −0.787               |
| 7/25/2003  | A                | 1            |                                                     | −7                            | −0.524               |
| 8/22/2003  | A                | 1            |                                                     | −9                            | −0.326               |
| 9/26/2003  | A                | 1            |                                                     | −6                            | 1.301                |
| 10/24/2003 | A                | 1            |                                                     | −10                           | −0.109               |
| 11/21/2003 | A                | 1            |                                                     | 2                             | 0.250                |
| 12/11/2003 | C                | 1            | Ortiz wins 2nd term as central bank governor        | 7                             | −0.868               |
| 1/23/2004  | A                | 1            |                                                     | 8                             | −1.615               |
| 2/20/2004  | A                | 1            |                                                     | 0                             | −0.067               |
| 3/12/2004  | C                | 1            | Central Bank announces change in daily USD auctions | −3                            | −0.653               |
| 3/26/2004  | A                | 1            |                                                     | −15                           | 0.640                |
| 4/23/2004  | A                | 1            |                                                     | −5                            | −0.128               |
| 4/27/2004  | A                | 1            |                                                     | −15                           | 0.002                |
| 5/28/2004  | C                | 1            | Bilateral treaty with Mercosur announced            | 2                             | −0.112               |
| 6/25/2004  | A                | 1            |                                                     | −5                            | 0.192                |
| 7/23/2004  | A                | 1            |                                                     | −2                            | 0.164                |
| 8/27/2004  | A                | 1            |                                                     | −4                            | 0.273                |
| 9/24/2004  | A                | 1            |                                                     | −3                            | 0.186                |
| 10/22/2004 | A                | 1            |                                                     | 1                             | 0.585                |
| 11/26/2004 | A                | 1            |                                                     | −3                            | −0.126               |
| 12/10/2004 | A                | 1            |                                                     | 1                             | 0.682                |
| 1/28/2005  | A                | 1            |                                                     | 2                             | −0.260               |
| 2/25/2005  | A                | 1            |                                                     | −2                            | −0.234               |
| 3/23/2005  | A                | 1            |                                                     | 4                             | 0.236                |
| 4/22/2005  | A                | 1            |                                                     | 1                             | 0.340                |
| 5/27/2005  | B                | 1            | Unemployment statistics released                    | 1                             | −0.691               |
| 6/24/2005  | B                | 1            | Unemployment statistics released                    | −2                            | −0.204               |
| 7/22/2005  | A                | 1            |                                                     | −3                            | 0.230                |
| 8/26/2005  | A                | 1            |                                                     | 0                             | 0.438                |
| 9/23/2005  | A                | 1            |                                                     | −2                            | 0.018                |
| 10/28/2005 | A                | 1            |                                                     | −2                            | −0.684               |
| 11/25/2005 | A                | 1            |                                                     | −1                            | −0.019               |
| 12/9/2005  | A                | 1            |                                                     | 2                             | 0.989                |
| 1/27/2006  | A                | 1            |                                                     | −3                            | −0.492               |
| 2/24/2006  | A                | 1            |                                                     | 0                             | −0.026               |
| 3/24/2006  | A                | 1            |                                                     | −1                            | −0.344               |
| 4/21/2006  | A                | 1            |                                                     | 0                             | 0.075                |
| 5/26/2006  | A                | 1            |                                                     | −1                            | −0.155               |
| 6/23/2006  | A                | 1            |                                                     | 1                             | −0.321               |
| 7/28/2006  | A                | 1            |                                                     | 0                             | −0.466               |
| 8/25/2006  | A                | 1            |                                                     | −1                            | 0.322                |
| 9/22/2006  | A                | 1            |                                                     | 0                             | −0.406               |
| 10/27/2006 | A                | 1            |                                                     | 0                             | 0.046                |
| 11/24/2006 | A                | 1            |                                                     | 1                             | 0.349                |
| 12/8/2006  | C                | 1            | Calderon presents first budget proposal             | 0                             | −0.478               |
| 1/26/2007  | A                | 1            |                                                     | 0                             | 0.208                |
| 2/23/2007  | A                | 1            |                                                     | 0                             | 0.336                |
| 3/23/2007  | A                | 1            |                                                     | 1                             | 0.016                |
| 4/27/2007  | A                | 1            |                                                     | 0                             | −0.261               |
| 5/25/2007  | A                | 1            |                                                     | 0                             | −0.487               |

Table 6 (continued)

| Meeting    | Exogeneity class | Intervention | Event                                              | $\Delta \Pi E28$ (b.p.) | $\Delta e$ (MXN/USD) |
|------------|------------------|--------------|----------------------------------------------------|-------------------------|----------------------|
| 6/22/2007  | A                | 1            |                                                    | 0                       | 0.239                |
| 7/27/2007  | A                | 1            |                                                    | 0                       | 0.362                |
| 8/24/2007  | B                | 1            | Unemployment statistics released                   | 0                       | –0.562               |
| 9/21/2007  | B                | 1            | Unemployment statistics released                   | –1                      | –0.305               |
| 10/26/2007 | A                | 1            |                                                    | 0                       | –0.731               |
| 11/23/2007 | A                | 1            |                                                    | 0                       | 0.055                |
| 12/7/2007  | C                | 1            | CPI statistics released                            | –2                      | –0.057               |
| 1/18/2008  | A                | 1            |                                                    | –1                      | –0.422               |
| 2/15/2008  | B                | 1            | Bombs near Mexico City police station              | 0                       | –0.007               |
| 3/14/2008  | A                | 1            |                                                    | –3                      | 0.143                |
| 4/18/2008  | B                | 1            | Unemployment statistics released                   | 1                       | 0.076                |
| 5/16/2008  | A                | 1            |                                                    | 2                       | –0.470               |
| 6/20/2008  | B                | 1            | Unemployment statistics released                   | 2                       | –0.345               |
| 7/18/2008  | B                | 1            | Unemployment statistics released                   | –2                      | –0.568               |
| 8/15/2008  | A                | 0            |                                                    | –2                      | 0.179                |
| 9/19/2008  | C                | 0            | US financial measures & Unemployment statistics    | 0                       | –0.826               |
| 10/17/2008 | A                | 0            |                                                    | –1                      | 0.325                |
| 11/28/2008 | A                | 0            |                                                    | 0                       | 1.974                |
| 1/16/2009  | A                | 0            |                                                    | –3                      | –0.264               |
| 2/20/2009  | C                | 1            | GDP statistics released                            | –4                      | 0.103                |
| 3/20/2009  | A                | 1            |                                                    | –6                      | –0.483               |
| 4/17/2009  | C                | 1            | IMF unveils \$47 bn flexible credit line to Mexico | –6                      | 0.299                |
| 5/15/2009  | B                | 1            | Unemployment statistics released                   | –6                      | 0.460                |
| 6/19/2009  | A                | 1            |                                                    | –2                      | –0.225               |
| 7/17/2009  | A                | 1            |                                                    | 2                       | –1.573               |
| 8/21/2009  | C                | 1            | GDP statistics released                            | –1                      | –0.414               |
| 9/18/2009  | A                | 1            |                                                    | –1                      | 0.003                |
| 10/16/2009 | A                | 0            |                                                    | –3                      | 0.228                |
| 11/27/2009 | A                | 0            |                                                    | 1                       | –0.533               |
| 1/15/2010  | A                | 0            |                                                    | 0                       | 0.197                |
| 2/19/2010  | A                | 0            |                                                    | 2                       | –0.590               |
| 3/19/2010  | A                | 0            |                                                    | 1                       | 0.495                |
| 4/16/2010  | A                | 0            |                                                    | 1                       | 0.896                |
| 5/21/2010  | A                | 0            |                                                    | 0                       | –1.496               |
| 6/18/2010  | A                | 0            |                                                    | –2                      | –0.612               |
| 7/16/2010  | B                | 0            | Federal police dies in car bombing                 | 1                       | 1.277                |
| 8/20/2010  | C                | 0            | GDP statistics released                            | 2                       | 0.474                |
| 9/24/2010  | A                | 0            |                                                    | –2                      | –0.929               |
| 10/15/2010 | A                | 0            |                                                    | –2                      | 0.176                |
| 11/26/2010 | A                | 0            |                                                    | 4                       | 0.848                |
| 1/21/2011  | B                | 0            | Unemployment statistics released                   | 2                       | 0.121                |
| 3/4/2011   | A                | 0            |                                                    | –1                      | 0.017                |
| 4/15/2011  | A                | 0            |                                                    | –3                      | –0.422               |
| 5/27/2011  | A                | 0            |                                                    | 1                       | –0.554               |

baskets of currencies were considered. Table 9 reports the test results when the basket included the US Dollar, the Euro, the Japanese Yen and the British Pound (which are the four currencies with highest turnover according to the Triennial Bank Survey of the *Bank of International Settlements*). The weight of each currency was calculated based on the share of total trade flows of each country with the respective currency areas during the 3 year period that ended in 2002.<sup>24</sup> The estimates of  $\beta$  change surprisingly little when compared to the estimates that are based on the US Dollar exchange rates. In Chile, a 100 b.p. increase in the base rate now leads to a 2.3–2.6% depreciation of the Peso in effective

<sup>24</sup> In the case of the 4 currency basket, for instance, the weights in the basket are 0.4366 for the US Dollar, 0.4336 for the Euro, 0.0817 for the Yen and 0.0481 for the Pound in the case of Brazil, 0.3436, 0.3280, 0.2236 and 0.1049, respectively, for Chile and 0.8950, 0.0674, 0.0307 and 0.0069, respectively, for Mexico.

**Table 7**

BCCh monetary policy meetings.

| Meeting    | Exogeneity class | Intervention | Event                                                          | $\Delta \text{swap}_{90}$<br>(b.p.) | $\Delta e$<br>(CLP/USD) |
|------------|------------------|--------------|----------------------------------------------------------------|-------------------------------------|-------------------------|
| 6/15/2006  | A                | 0            |                                                                | 0                                   | 0.046                   |
| 7/13/2006  | A                | 0            |                                                                | 0                                   | −0.064                  |
| 8/10/2006  | A                | 0            |                                                                | 0                                   | −0.050                  |
| 9/7/2006   | A                | 0            |                                                                | −1                                  | 0.108                   |
| 10/12/2006 | A                | 0            |                                                                | 0                                   | −0.235                  |
| 11/16/2006 | A                | 0            |                                                                | −3                                  | 0.398                   |
| 12/14/2006 | C                | 0            | S&P raises long term foreign currency outlook                  | −3                                  | −0.156                  |
| 1/11/2007  | A                | 0            |                                                                | −20                                 | 0.471                   |
| 2/8/2007   | A                | 0            |                                                                | 0                                   | −0.521                  |
| 3/15/2007  | A                | 0            |                                                                | 1                                   | −0.380                  |
| 4/12/2007  | A                | 0            |                                                                | 3                                   | 0.023                   |
| 5/10/2007  | A                | 0            |                                                                | 1                                   | −0.348                  |
| 6/14/2007  | A                | 0            |                                                                | 1                                   | −0.588                  |
| 7/12/2007  | A                | 0            |                                                                | 17                                  | −0.149                  |
| 8/9/2007   | C                | 0            | CBs in US, Eurozone, Japan inject money into banking system    | 16                                  | 0.258                   |
| 9/13/2007  | A                | 0            |                                                                | −6                                  | 0.345                   |
| 10/11/2007 | A                | 0            |                                                                | −2                                  | 0.117                   |
| 11/13/2007 | C                | 0            | 7.7 magnitude quake hits northern Chile                        | 0                                   | −0.471                  |
| 12/13/2007 | A                | 0            |                                                                | 21                                  | −0.284                  |
| 1/10/2008  | A                | 0            |                                                                | 31                                  | −1.382                  |
| 2/7/2008   | A                | 0            |                                                                | −1                                  | −0.890                  |
| 3/13/2008  | A                | 0            |                                                                | −2                                  | 1.197                   |
| 4/10/2008  | C                | 0            | CB announces US Dollars purchase program                       | −3                                  | 3.025                   |
| 5/8/2008   | A                | 1            |                                                                | −1                                  | −0.710                  |
| 6/10/2008  | A                | 1            |                                                                | 31                                  | 0.893                   |
| 7/10/2008  | A                | 1            |                                                                | 46                                  | 0.380                   |
| 8/14/2008  | A                | 1            |                                                                | 2                                   | 0.856                   |
| 9/4/2008   | A                | 1            |                                                                | 5                                   | 1.726                   |
| 10/9/2008  | C                | 0            | BCCh extends its US Dollar swaps program to \$ 5bn             | −8                                  | 4.431                   |
| 11/13/2008 | B                | 0            | Government workers reject salary offer in national strike      | −15                                 | −0.437                  |
| 12/11/2008 | A                | 0            |                                                                | −3                                  | −1.384                  |
| 1/8/2009   | A                | 0            |                                                                | −40                                 | −1.798                  |
| 2/12/2009  | A                | 0            |                                                                | −102                                | −2.390                  |
| 3/12/2009  | A                | 0            |                                                                | −21                                 | −0.527                  |
| 4/9/2009   | A                | 0            |                                                                | 0                                   | −0.208                  |
| 5/7/2009   | A                | 0            |                                                                | −18                                 | −0.088                  |
| 6/16/2009  | C                | 0            | CB announces US Dollars sale program                           | 0                                   | −1.611                  |
| 7/9/2009   | A                | 0            |                                                                | −1                                  | 0.320                   |
| 8/13/2009  | A                | 0            |                                                                | 1                                   | 0.506                   |
| 9/8/2009   | A                | 0            |                                                                | 1                                   | 0.018                   |
| 10/13/2009 | A                | 0            |                                                                | 0                                   | −0.665                  |
| 11/12/2009 | A                | 0            |                                                                | 0                                   | −1.064                  |
| 12/15/2009 | B                | 0            | Formal invitation to join OECD                                 | −2                                  | −0.190                  |
| 1/14/2010  | A                | 0            |                                                                | 0                                   | −0.902                  |
| 2/11/2010  | A                | 0            |                                                                | 0                                   | −0.770                  |
| 3/18/2010  | B                | 0            | Chile considers new tax to fund reconstruction                 | 0                                   | 0.609                   |
| 4/15/2010  | B                | 0            | Chile considers raising corporate tax by 3% in 2011            | 1                                   | 0.576                   |
| 5/13/2010  | A                | 0            |                                                                | −20                                 | 1.133                   |
| 6/15/2010  | C                | 0            | Senate rejects change to mining royalty bill & Moody's upgrade | 23                                  | −0.953                  |
| 7/15/2010  | A                | 0            |                                                                | 4                                   | 0.038                   |
| 8/12/2010  | A                | 0            |                                                                | 8                                   | −0.024                  |
| 9/16/2010  | A                | 0            |                                                                | 0                                   | −0.140                  |
| 10/14/2010 | A                | 0            |                                                                | −7                                  | 0.065                   |
| 11/15/2010 | A                | 0            |                                                                | 2                                   | −0.391                  |
| 12/16/2010 | A                | 0            |                                                                | −10                                 | −0.254                  |

Table 7 (continued)

| Meeting   | Exogeneity class | Intervention | Event                   | $\Delta \text{swap}_{90}$<br>(b.p.) | $\Delta e$<br>(CLP/USD) |
|-----------|------------------|--------------|-------------------------|-------------------------------------|-------------------------|
| 1/13/2011 | A                | 1            |                         | –1                                  | –0.102                  |
| 2/17/2011 | A                | 1            |                         | 5                                   | –0.238                  |
| 3/17/2011 | C                | 1            | GDP statistics released | 45                                  | –1.059                  |
| 4/12/2011 | A                | 1            |                         | 9                                   | –0.359                  |
| 5/12/2011 | A                | 1            |                         | 23                                  | 0.259                   |

Table 8

The impact of monetary policy on the exchange rate – by classes of exogeneity.

| Exogeneity classes              | All             |                 | Excluding C     |                 | Excluding B & C |                 |
|---------------------------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|
|                                 | L.S.            |                 | L.S.            |                 | L.S.            |                 |
|                                 | All             | No interv.      | All             | No interv.      | All             | No interv.      |
| Brazil (change in BRL/USD rate) |                 |                 |                 |                 |                 |                 |
| <b>d (DI 90)</b>                | <b>1.368</b>    | <b>1.599</b>    | <b>1.307</b>    | <b>1.524</b>    | <b>1.133</b>    | <b>1.523</b>    |
|                                 | <b>0.825</b>    | <b>1.186</b>    | <b>0.848</b>    | <b>1.217</b>    | <b>0.839</b>    | <b>1.224</b>    |
| d (3 month T Bill)              | 0.672           | 3.764           | 2.750           | 8.687***        | 5.284***        | 8.551***        |
|                                 | 2.733           | 3.297           | 2.790           | 2.448           | 1.920           | 2.508           |
| d (VIX)                         | 0.334***        | 0.298***        | 0.423***        | 0.525***        | 0.438***        | 0.531***        |
|                                 | 0.088           | 0.109           | 0.086           | 0.123           | 0.087           | 0.121           |
| d (price of commodities)        | 0.068           | 0.365           | –0.068          | 0.199           | 0.040           | 0.260           |
|                                 | 0.232           | 0.255           | 0.238           | 0.255           | 0.199           | 0.259           |
| No. of observations             | 79              | 38              | 70              | 33              | 62              | 32              |
| R2                              | 0.210           | 0.293           | 0.251           | 0.405           | 0.363           | 0.426           |
| RMSE                            | 0.920           | 0.846           | 0.889           | 0.748           | 0.759           | 0.743           |
| F/chi2                          | 4.42***         | 3.66**          | 6.69***         | 5.43***         | 7.52***         | 5.76***         |
| Mexico (change in MXN/USD rate) |                 |                 |                 |                 |                 |                 |
| <b>d (TIE 28)</b>               | <b>–0.439</b>   | <b>1.192</b>    | <b>0.049</b>    | <b>1.387</b>    | <b>0.034</b>    | <b>0.904</b>    |
|                                 | <b>1.086</b>    | <b>1.464</b>    | <b>1.042</b>    | <b>1.669</b>    | <b>1.050</b>    | <b>1.811</b>    |
| d (1 m USD LIBOR)               | 1.711*          | 1.603*          | 1.704*          | 1.612*          | 1.683*          | 1.628*          |
|                                 | 0.874           | 0.855           | 0.872           | 0.848           | 0.880           | 0.870           |
| d (VIX)                         | 0.163***        | 0.164*          | 0.162***        | 0.160           | 0.152**         | 0.154           |
|                                 | 0.053           | 0.093           | 0.057           | 0.094           | 0.060           | 0.091           |
| d (price of WTI crude oil)      | –0.030          | –0.032          | –0.029          | –0.011          | –0.028          | 0.007           |
|                                 | 0.018           | 0.057           | 0.019           | 0.076           | 0.019           | 0.080           |
| No. of observations             | 101             | 28 (small N)    | 89              | 24 (small N)    | 78              | 22 (small N)    |
| R2                              | 0.240           | 0.336           | 0.241           | 0.356           | 0.227           | 0.373           |
| RMSE                            | 0.521           | 0.686           | 0.526           | 0.697           | 0.539           | 0.692           |
| F/chi2                          | 3.39**          | 1.55            | 2.87**          | 1.99            | 2.26*           | 1.82            |
| Exogeneity classes              | All             |                 | Excluding C     |                 | Excluding B & C |                 |
|                                 | I.V.            |                 | I.V.            |                 | I.V.            |                 |
|                                 | All             | No interv.      | All             | No interv.      | All             | No interv.      |
| Chile (change in CLP/USD rate)  |                 |                 |                 |                 |                 |                 |
| <b>d (swap 90)</b>              | <b>2.206***</b> | <b>2.296***</b> | <b>2.455***</b> | <b>2.401***</b> | <b>2.510***</b> | <b>2.525***</b> |
|                                 | <b>0.439</b>    | <b>0.443</b>    | <b>0.330</b>    | <b>0.290</b>    | <b>0.361</b>    | <b>0.330</b>    |
| d (3 month T Bill)              | –2.592          | –3.999          | –0.896          | –2.831          | –0.630          | –2.556          |
|                                 | 2.591           | 2.797           | 2.941           | 2.284           | 2.920           | 2.197           |
| d (VIX)                         | 0.200**         | 0.185*          | 0.111           | 0.111           | 0.161*          | 0.181*          |
|                                 | 0.089           | 0.096           | 0.077           | 0.086           | 0.097           | 0.102           |
| d (price of commodities)        | –0.493***       | –0.377**        | –0.210          | –0.102          | –0.210          | –0.081          |
|                                 | 0.170           | 0.170           | 0.155           | 0.159           | 0.150           | 0.141           |
| No. of observations             | 58              | 48              | 50              | 41              | 46              | 37              |
| R2                              | 0.434           | 0.469           | 0.318           | 0.358           | 0.330           | 0.386           |
| RMSE                            | 0.764           | 0.766           | 0.589           | 0.544           | 0.595           | 0.539           |
| F/chi2                          | 49.33***        | 46.22***        | 76.83***        | 108.09***       | 70.86***        | 99.85***        |

Note: Robust standard errors are reported. \*, \*\* and \*\*\* denote statistical significance at the 10%, 5% and 1% confidence levels, respectively.

**Table 9**

The impact of monetary policy on the effective exchange rate (basket of 4 currencies).

| Exogeneity classes              | All             |                             | Excluding C     |                             | Excluding B & C |                             |
|---------------------------------|-----------------|-----------------------------|-----------------|-----------------------------|-----------------|-----------------------------|
|                                 | L.S.            |                             | L.S.            |                             | L.S.            |                             |
|                                 | All             | No interv.                  | All             | No interv.                  | All             | No interv.                  |
| Brazil (change in BRL/USD rate) |                 |                             |                 |                             |                 |                             |
| <b>d (DI 90)</b>                | <b>1.217</b>    | <b>1.327</b>                | <b>1.174</b>    | <b>1.260</b>                | <b>1.101</b>    | <b>1.258</b>                |
|                                 | <b>0.769</b>    | <b>1.170</b>                | <b>0.795</b>    | <b>1.225</b>                | <b>0.797</b>    | <b>1.232</b>                |
| d (3 month T Bill)              | 0.713           | 4.055                       | 2.432           | 8.516***                    | 4.361**         | 8.391***                    |
|                                 | 2.344           | 2.820                       | 2.421           | 2.139                       | 1.939           | 2.183                       |
| d (VIX)                         | 0.335***        | 0.304***                    | 0.408***        | 0.490***                    | 0.422***        | 0.496***                    |
|                                 | 0.078           | 0.100                       | 0.080           | 0.139                       | 0.083           | 0.138                       |
| d (price of commodities)        | 0.072           | 0.329                       | −0.019          | 0.209                       | −0.002          | 0.269                       |
|                                 | 0.198           | 0.256                       | 0.203           | 0.279                       | 0.201           | 0.284                       |
| No. of observations             | 79              | 38                          | 70              | 33                          | 62              | 32                          |
| R2                              | 0.237           | 0.278                       | 0.264           | 0.336                       | 0.331           | 0.354                       |
| RMSE                            | 0.839           | 0.865                       | 0.814           | 0.799                       | 0.767           | 0.797                       |
| F/chi2                          | 6.20***         | 4.27***                     | 7.03***         | 4.87***                     | 7.20***         | 4.91***                     |
| Mexico (change in MXN/USD rate) |                 |                             |                 |                             |                 |                             |
| <b>d (TIE 28)</b>               | <b>−0.539</b>   | <b>1.130</b>                | <b>−0.065</b>   | <b>1.295</b>                | <b>−0.118</b>   | <b>0.763</b>                |
|                                 | <b>1.108</b>    | <b>1.483</b>                | <b>1.077</b>    | <b>1.705</b>                | <b>1.089</b>    | <b>1.853</b>                |
| d (1 m USD LIBOR)               | 1.616**         | 1.510*                      | 1.604**         | 1.516*                      | 1.588*          | 1.535*                      |
|                                 | 0.803           | 0.775                       | 0.803           | 0.755                       | 0.813           | 0.778                       |
| d (VIX)                         | 0.164***        | 0.163*                      | 0.160***        | 0.157*                      | 0.150**         | 0.151                       |
|                                 | 0.051           | 0.090                       | 0.055           | 0.091                       | 0.058           | 0.088                       |
| d (price of WTI crude oil)      | −0.025          | −0.029                      | −0.025          | −0.005                      | −0.016          | 0.013                       |
|                                 | 0.018           | 0.057                       | 0.018           | 0.075                       | 0.019           | 0.080                       |
| No. of observations             | 101             | 28 <small>(small N)</small> | 89              | 24 <small>(small N)</small> | 78              | 22 <small>(small N)</small> |
| R2                              | 0.237           | 0.337                       | 0.231           | 0.355                       | 0.217           | 0.376                       |
| RMSE                            | 0.518           | 0.664                       | 0.526           | 0.675                       | 0.539           | 0.666                       |
| F/chi2                          | 3.48**          | 1.63                        | 2.87**          | 2.17                        | 2.33*           | 2.00                        |
| Exogeneity classes              | All             |                             | Excluding C     |                             | Excluding B & C |                             |
|                                 | I.V.            |                             | I.V.            |                             | I.V.            |                             |
|                                 | All             | No interv.                  | All             | No interv.                  | All             | No interv.                  |
| Chile (change in CLP/USD rate)  |                 |                             |                 |                             |                 |                             |
| <b>d (swap 90)</b>              | <b>2.308***</b> | <b>2.492***</b>             | <b>2.485***</b> | <b>2.555***</b>             | <b>2.518***</b> | <b>2.638***</b>             |
|                                 | <b>0.484</b>    | <b>0.522</b>                | <b>0.384</b>    | <b>0.365</b>                | <b>0.410</b>    | <b>0.398</b>                |
| d (3 month T Bill)              | −1.514          | −2.630                      | 1.017           | −1.268                      | 1.148           | −1.056                      |
|                                 | 2.701           | 2.889                       | 2.847           | 2.087                       | 2.952           | 2.321                       |
| d (VIX)                         | 0.258***        | 0.254***                    | 0.172**         | 0.175**                     | 0.210*          | 0.237**                     |
|                                 | 0.085           | 0.093                       | 0.076           | 0.082                       | 0.108           | 0.107                       |
| d (price of commodities)        | −0.646***       | −0.578***                   | −0.349*         | −0.293*                     | −0.353*         | −0.276*                     |
|                                 | 0.192           | 0.176                       | 0.186           | 0.174                       | 0.182           | 0.161                       |
| No. of observations             | 58              | 48                          | 50              | 41                          | 46              | 37                          |
| R2                              | 0.471           | 0.524                       | 0.286           | 0.358                       | 0.293           | 0.372                       |
| RMSE                            | 0.805           | 0.791                       | 0.662           | 0.601                       | 0.665           | 0.595                       |
| F/chi2                          | 45.05***        | 49.47***                    | 47.40***        | 59.98***                    | 43.86***        | 59.46***                    |

Note: Robust standard errors are reported. \*, \*\* and \*\*\* denote statistical significance at the 10%, 5% and 1% confidence levels, respectively.

terms – a result that continues to be significant at the 1% confidence level and is practically identical to that obtained based on bilateral rates.<sup>25</sup>

<sup>25</sup> Using copper price instead of the CRB for Chile does not change this result either. We continue to find that an interest rate hike leads to a depreciation at the 1% confidence level in all 14 specifications we use (in Tables 1, 5 and 6). The estimated magnitude of the effect varies between 1.7% and 2.4%. This is hardly surprising as the correlation between the variations in the CRB and copper prices is 0.517. The substitution of the CRB by copper price however causes the control for foreign monetary conditions to attain the wrong sign, with significance, in some cases.

**Table 10**

Testing for non-linearities.

|                                          | Full sample |           | No intervention |              |
|------------------------------------------|-------------|-----------|-----------------|--------------|
| D.V.: change in BRL/USD rate             |             |           |                 |              |
| Brazil                                   |             |           |                 |              |
| d (DI 90)                                | 0.866       | 1.884*    | 2.433           | 2.592*       |
|                                          | 1.234       | 1.127     | 1.567           | 1.418        |
| d (DI 90)* dummy (var>90th percentile)   | 0.989       | −0.776    | −1.466          | −1.549       |
|                                          | 0.827       | 1.506     | 1.900           | 1.721        |
| Control variables                        | No          | Yes       | No              | Yes          |
| No. of observations                      | 79          | 79        | 38              | 38           |
| R2                                       | 0.025       | 0.213     | 0.082           | 0.317        |
| RMSE                                     | 1.009       | 0.924     | 0.948           | 0.855        |
| F/chi2                                   | 0.72        | 3.58***   | 1.57            | 2.97**       |
| D.V.: change in MXN/USD rate             |             |           |                 |              |
| Mexico                                   |             |           |                 |              |
| d (TIE 28)                               | 1.147       | 0.607     | 10.898*         | 9.131        |
|                                          | 2.954       | 2.841     | 6.024           | 5.387        |
| d (TIE 28)* dummy (var>90th percentile)  | −2.089      | −1.277    | −11.376*        | −9.219       |
|                                          | 2.955       | 2.928     | 6.337           | 5.852        |
| Control variables                        | no          | yes       | no              | yes          |
| No. of observations                      | 101         | 101       | 28 (small N)    | 28 (small N) |
| R2                                       | 0.009       | 0.242     | 0.118           | 0.403        |
| RMSE                                     | 0.588       | 0.530     | 0.758           | 0.665        |
| F/chi2                                   | 0.54        | 2.89**    | 1.67            | 2.97**       |
| D.V.: change in CLP/USD rate             |             |           |                 |              |
| Chile (I.V.)                             |             |           |                 |              |
| d (swap 90)                              | 2.518**     | 2.507*    | 1.658           | 1.825        |
|                                          | 1.221       | 1.393     | 1.394           | 1.687        |
| d (swap 90)* dummy (var>90th percentile) | −0.356      | −0.410    | 0.993           | 0.602        |
|                                          | 1.315       | 1.391     | 1.423           | 1.660        |
| Control variables                        | No          | Yes       | No              | Yes          |
| No. of observations                      | 58          | 58        | 48              | 48           |
| R2                                       | 0.004       | 0.422     | 0.082           | 0.491        |
| RMSE                                     | 1.019       | 0.771     | 1.007           | 0.750        |
| F/chi2                                   | 28.39***    | 104.41*** | 45.02***        | 200.36***    |

Note: Robust standard errors. \*, \*\* and \*\*\* denote statistical significance at the 10%, 5% and 1% confidence levels, respectively.

### 6.3. Testing for Non-Linearities

A number of studies have noted that the relationship between interest rates and exchange rates may actually be non-linear. This could be due to the existence of transaction costs or limits to speculation. Sarno et al. (2006), for instance, argue that investors would only want to take up a currency strategy if the strategy had a Sharpe ratio above that of alternative investments. This would imply that there could be a band of inaction where inefficiencies are too small to attract speculative capital. These types of arguments typically suggest that there could be non-linear adjustments to changes in the interest rates, i.e., the exchange rate would be more likely to move following larger changes in the base rate, as changes of larger magnitude would be more likely to push the investor out of his band of inaction.

In order to check if there is indicative evidence of whether non-linearities could explain the results of the previous sections, we altered the baseline specification in (1) to

$$\Delta e_t = \alpha + \beta \Delta i_{n,t} + \delta \Delta i_{n,t} D + \gamma \Delta Z_t + \varepsilon_t \quad (2)$$

where  $D$  represents a large shock dummy variable. More specifically, this variable indicates whether the absolute value of the rate change lies within the top decile of variations that were observed for each country during the sample period. A significant  $\delta$  coefficient would be a clear indication of a non-linear

response pattern. Note that we purposely focus on the effects of domestic monetary policy events, as these are the source of the more puzzling results of Section 4.<sup>26</sup>

Table 10 shows the results for these estimations for each country. In the case of Chile, the Durbin score and the Wu–Hausman statistic continued to reject the hypothesis of exogeneity of the regressors, which is why we proceed reporting the results that were based on I.V. estimations instead. This was not the case for Brazil and Mexico, where we used OLS, as before. While the sign of the coefficient of the interaction term is indeed negative in 9 out of the 12 cases, this term is generally far from being statistically significant.<sup>27</sup>

To gauge the effect of the large monetary policy shocks the two coefficients that are reported in the table need to be added. Statistical tests clearly do not allow us to reject the rather stark hypothesis that the effect of large monetary policy shocks on the exchange is zero in the cases of Brazil and Mexico (with *p*-values that vary between 0.30 and 0.97). Only in the case of Chile was this null rejected. However, the effects of large shocks in Chile were always found to be positive rather than negative, with the sum of coefficients varying between 2.1 and 2.7%.<sup>28</sup>

We also made use of plain and risk adjusted measures of deviations from uncovered interest parity to construct large UIP deviation dummy variables. These alternative dummies flag all the cases where the absolute value of each deviation measure lies within the top decile of each sample. Equation (2) was then re-estimated with these large UIP deviation dummy variables replacing the large shock dummy that we had used previously. To construct the measures of deviation from UIP, we used the mean expected rates of depreciation of the Brazilian Real that chief economists report in the Brazilian Central Bank survey (for a more detailed description of this survey, see Carvalho and Minella (2012)).<sup>29</sup>

The first deviation dummy was simply based on the absolute value of the interest rate differential (i.e. SELIC – Fed Funds rate) that prevailed at the day of each COPOM meeting, ignoring expected changes in the exchange rate.<sup>30</sup> The second estimation used the dummy that was constructed based on the deviation measure that corrected the interest rate differential for the expected rate of depreciation of the Real over the next 12 months – as captured by the survey of expectations. We then adjusted these first two measures of expected returns using a proxy for return volatility. More specifically, we used the ratio between the two measures of expected returns and the implicit at the money USDBRL option volatility on the day of each meeting.<sup>31</sup> The time series for these two ratios were then used to construct our last two large UIP deviation dummies. Interestingly, the large deviation meetings did not coincide with any period in which there was positive taxation of portfolio inflows.<sup>32</sup> Table A1 in the Appendix reports the results of the estimation of Equation (2) when each of the four alternative large deviation dummies were used. While the  $\delta$  coefficient is negative in 9 of the 16 cases, it did not turn out to be significant in any of the 16 cases. Moreover, the sum of the two estimated  $d$  ( $DI90$ ) coefficients is positive in 15 out of 16 cases, rather than negative (though only statistically significant at 10% once).<sup>33</sup> This suggests that it would be hard to ascertain that the exchange rate effects of monetary policy shocks that occur at times when there are large absolute deviations from interest parity are different than those of usual monetary shocks.

<sup>26</sup> Furthermore, these are also the larger surprises.

<sup>27</sup> With the exception of one case for Mexico, where  $\delta$  was found to be significant at 10%. An important caveat, however, is that this particular estimation (for the sample without interventions) is based on a very small number of degrees of freedom.

<sup>28</sup> It is important to note that, as one would expect, the non-linear specification leads to much larger standard errors of the coefficients, compared to the linear specifications of Table 1.

<sup>29</sup> On the Friday immediately prior to the monetary policy event.

<sup>30</sup> Or, alternatively, assuming that the expected rate of depreciation is zero (i.e. the exchange rate follows a random walk without drift).

<sup>31</sup> The source of this last series is Bloomberg.

<sup>32</sup> This is partly due to the fact that the IOF tax – which was positive between March and October 2008 and again between October 2009 and June 2013 – was active during a period when the interest rate differential was actually well below the recent historical standards. Incidentally, only the second deviation dummy picked up events that occurred after the first introduction of the IOF tax. These are two events that occurred during the global financial crisis, when the tax was at zero.

<sup>33</sup> The null of no effect of large deviation shocks is not rejected at 10% in the only case where the sum of the coefficients is negative.

All in all, we do not find that the effects of large monetary policy shocks, or shocks that are associated with large deviations from interest parity, are significantly different (i.e.  $\delta$  is not significant). One should add the important point that it would appear difficult to reconcile the non-linearity hypothesis with the fact that we do find the expected exchange rate responses to U.S. monetary policy changes in the cases of Brazil and Mexico.

#### 6.4. Using survey data

In principle, one could still question whether 1-day market interest rate changes or actual base rate changes are the best available *proxies* for target rate surprises. While we do believe that these are reasonable *proxies* and the choice follows the practice in the literature (see for instance the arguments for using them in Kuttner (2001), Zettelmeyer (2004), Kearns and Manners (2006) and Faust et al. (2007)), we were able to re-estimate the regressions in Table 1 using a monetary surprise *proxy* that is based on market survey data in the case of Brazil. For this, we made use of the Central Bank survey that is described in Carvalho and Minella (2012) again. The monetary innovation regressor was constructed by taking the difference between the new target rate and the mean rate that was expected by market participants on the last Friday prior to the meeting (i.e. three working days before decision and the announcement). Table A2 in the Appendix shows the results that were obtained when this covariate was used instead. On the whole, there is no qualitative change, even though R2s get a bit worse, when compared to those of Table 1. The main coefficient of interest remains positive in all specifications, but is far from statistical significance. The only variable that is clearly significant, as before, is the VIX.

#### 6.5. Using direct country risk controls

In order to minimize the possibility of reverse causality we had used the change in the VIX as a proxy for country risk, noting the fact that the sample correlation between the VIX and the 5 year CDS spreads was above 0.5 in all three countries. As a robustness check, we now include the change in the country specific CDS spreads among the control variables directly. For each country we include first the 5 year CDS spread as the sole risk factor, and then in conjunction with the VIX. The results of this exercise can be seen in Table 11.

On the whole, the estimates of our main variable of interest  $\beta$  are not greatly affected by this change in the risk proxy. The qualitative results remain the same. The coefficient remains positive in all specifications in the samples without intervention, being significant for Chile. The substitution of the VIX rate by the CDS spreads however caused the control for the foreign monetary conditions to attain the wrong sign in the case of Chile.

#### 6.6. Checking for asymmetric responses

Finally, we also checked whether we could find any indication of asymmetry in the exchange rate response of positive and negative rate surprises. Table 12 reports the estimated  $\beta$  coefficients for each subsample for all specifications. We can never reject the hypothesis that the slope coefficients are equal for positive and negative surprises. This could be taken as further evidence against a fiscal explanation for the puzzle, as there does not seem to be any indication of the non-linearities that would be expected if we had a fiscal explanation.

#### 6.7. Effects on exchange rate expectations

We have also checked for evidence of whether monetary policy shocks produced any significant changes in the expected rate of depreciation of a currency.<sup>34</sup> Standard theory would suggest that, for instance, an increase in the rate differential would lead to expectations of greater depreciation of the home currency. To test this hypothesis, we made use of the Brazilian Central Bank survey again. This

<sup>34</sup> We thank an anonymous referee for making this suggestion.

**Table 11**

The impact of monetary policy on the exchange rate (CDS).

|                              | Full sample |          | No intervention |              |
|------------------------------|-------------|----------|-----------------|--------------|
|                              | L.S.        | L.S.     | L.S.            | L.S.         |
| D.V.: Change in BRL/USD rate |             |          |                 |              |
| Brazil                       |             |          |                 |              |
| d (DI 90)                    | 1.090       | 1.319*   | 1.340           | 1.476        |
|                              | 0.767       | 0.781    | 1.164           | 1.149        |
| d (3 month T Bill)           | 0.960       | 0.871    | 5.220***        | 4.117        |
|                              | 2.425       | 2.727    | 1.668           | 3.112        |
| d (CDS)                      | 2.097***    | 1.386*** | 1.512***        | 1.043***     |
|                              | 0.570       | 0.421    | 0.431           | 0.331        |
| d (VIX)                      |             | 0.279*** |                 | 0.251**      |
|                              |             | 0.091    |                 | 0.109        |
| d (price of commodities)     | -0.175      | 0.062    | 0.187           | 0.377        |
|                              | 0.209       | 0.224    | 0.217           | 0.232        |
| No. of observations          | 79          | 79       | 38              | 38           |
| R2                           | 0.146       | 0.251    | 0.251           | 0.349        |
| RMSE                         | 0.957       | 0.903    | 0.882           | 0.835        |
| F/chi2                       | 4.30***     | 4.70***  | 7.62***         | 3.92***      |
| D.V.: Change in MXN/USD rate |             |          |                 |              |
| Mexico                       |             |          |                 |              |
| d (TIE 28)                   | -0.716      | -0.537   | 0.428           | 0.593        |
|                              | 1.060       | 1.019    | 1.264           | 1.156        |
| d (1 m USD LIBOR)            | 1.550       | 1.702**  | 1.494           | 1.650*       |
|                              | 1.208       | 0.815    | 1.222           | 0.875        |
| d (CDS)                      | 0.836       | -0.049   | 1.859           | 1.206        |
|                              | 0.939       | 1.075    | 1.603           | 1.991        |
| d (VIX)                      |             | 0.161*** |                 | 0.158*       |
|                              |             | 0.054    |                 | 0.087        |
| d (price of commodities)     | -0.105      | -0.007   | -0.075          | 0.100        |
|                              | 0.100       | 0.119    | 0.306           | 0.390        |
| No. of observations          | 101         | 101      | 28 (small N)    | 28 (small N) |
| R2                           | 0.084       | 0.226    | 0.172           | 0.341        |
| RMSE                         | 0.572       | 0.528    | 0.766           | 0.699        |
| F/chi2                       | 1.15        | 2.55**   | 1.09            | 1.43         |
| D.V.: change in CLP/USD rate |             |          |                 |              |
|                              | Full Sample |          | No Intervention |              |
|                              | I.V.        | I.V.     | I.V.            | I.V.         |
| Chile                        |             |          |                 |              |
| d (swap 90)                  | 1.705***    | 1.987*** | 1.716***        | 1.988***     |
|                              | 0.450       | 0.492    | 0.506           | 0.531        |
| d (3 month T Bill)           | -4.574**    | -2.024   | -6.053***       | -3.698       |
|                              | 2.111       | 2.302    | 2.169           | 2.383        |
| d (CDS)                      | 7.563**     | 6.341*   | 8.893***        | 7.656**      |
|                              | 3.416       | 3.453    | 3.289           | 3.252        |
| d (VIX)                      |             | 0.179*   |                 | 0.156        |
|                              |             | 0.093    |                 | 0.098        |
| d (price of commodities)     | -0.330*     | -0.355** | -0.102          | -0.149       |
|                              | 0.186       | 0.169    | 0.170           | 0.145        |
| No. of observations          | 58          | 58       | 48              | 48           |
| R2                           | 0.436       | 0.472    | 0.486           | 0.517        |
| RMSE                         | 0.763       | 0.737    | 0.754           | 0.730        |
| F/chi2                       | 59.59***    | 60.11*** | 56.88***        | 57.53***     |

Note: Robust standard errors. \*, \*\* and \*\*\* denote statistical significance at the 10%, 5% and 1% confidence levels, respectively.

survey contains information on the mean USDBRL rate that chief economists expect to prevail at the end of each month. We used this information to construct a variable that measured the expected change in the USDBRL rate relative to the going market rate. The change in this variable during the weeks of the committee meetings was then taken as the dependent variable,<sup>35</sup> which was then

<sup>35</sup> Friday after vs. Friday before the meeting.

**Table 12**

Testing for asymmetries.

|                                 | $\beta$ estimate   | $\beta$ estimate   | Difference t-stat |
|---------------------------------|--------------------|--------------------|-------------------|
|                                 | Negative surprises | Positive surprises |                   |
| Brazil                          |                    |                    |                   |
| Full sample – w/o controls      | 0.956 (1.187)      | 1.931 (1.885)      | 0.438             |
| Full sample – with controls     | 1.053 (0.768)      | 1.940 (2.058)      | 0.404             |
| No intervention – w/o controls  | 0.481 (1.147)      | 1.253 (2.808)      | 0.254             |
| No intervention – with controls | 1.699 (0.970)      | 1.167 (2.772)      | –0.181            |
| Mexico                          |                    |                    |                   |
| Full sample – w/o controls      | –0.658 (0.896)     | –4.766 (9.007)     | –0.454            |
| Full sample – with controls     | –0.457 (0.839)     | –4.291 (8.196)     | –0.465            |
| No intervention – w/o controls  | Small N            | Small N            |                   |
| No intervention – with controls | Small N            | Small N            |                   |
| Chile                           |                    |                    |                   |
| Full sample – w/o controls      | 3.017 (0.631)      | 3.499 (2.205)      | 0.210             |
| Full sample – with controls     | 2.323 (0.536)      | 2.246 (1.692)      | –0.043            |
| No intervention – w/o controls  | 3.082 (0.663)      | 3.850 (4.596)      | 0.165             |
| No intervention – with controls | 2.313 (0.540)      | 0.276 (3.537)      | –0.569            |

Note: Robust standard errors in parenthesis.

**Table 13**

The impact of monetary policy on the expected rate of depreciation.

| D.V.: change in expected depreciation rate | Full sample | No intervention |
|--------------------------------------------|-------------|-----------------|
| 6 months horizon                           |             |                 |
| Constant                                   | –0.069      | –0.159          |
|                                            | 0.148       | 0.210           |
| d (DI 90)                                  | –0.832      | –1.160          |
|                                            | 0.868       | 1.321           |
| No. of observations                        | 79          | 38              |
| R2                                         | 0.011       | 0.022           |
| RMSE                                       | 1.301       | 1.356           |
| F/chi2                                     | 0.92        | 0.77            |
| 12 months horizon                          |             |                 |
| Constant                                   | –0.068      | –0.149          |
|                                            | 0.134       | 0.190           |
| d (TIIE 28)                                | –0.526      | –0.766          |
|                                            | 0.806       | 1.061           |
| No. of observations                        | 79          | 38              |
| R2                                         | 0.005       | 0.013           |
| RMSE                                       | 1.195       | 1.180           |
| F/chi2                                     | 0.43        | 0.52            |

Note: Robust standard errors. \*, \*\* and \*\*\* denote statistical significance at the 10%, 5% and 1% confidence levels, respectively.

regressed against the proxy for the monetary policy surprise. 6 and 12 month forecast horizons were used. The results of this exercise can be seen in Table 13. It is apparent that the monetary policy surprise proxy fails to explain changes in the expected rate of depreciation of the Brazilian Real. As with actual exchange rate variations in the case of Brazil, the effects of the changes in the market interest rate on exchange rate expectations are never significant.<sup>36</sup>

The absence of significant changes in the expected depreciation rates following monetary policy decisions is probably less of a surprise if we consider that we had already found no significant effect on the actual market rate, and that the expected exchange rates tend to track the spot exchange rate quite closely.

<sup>36</sup> Note also that the sign of the coefficient is actually negative, indicating an association of surprise hikes with expectations of appreciation rather than depreciation. The *p*-value for the hypothesis that this effect is zero, however, is always above 1/3.

## 7. Relation to the literature

There are some results in the literature that could be related to our findings. One paper that finds results that are similar to the ones reported here is that of [Kraay \(2003\)](#), who fails to find any indication that higher interest rates defend exchange rates that come under attack. However, their sample is restricted to periods of crises. In terms of theory, a number of models have weakened or even reversed the link between interest rate hikes and currency appreciations. In the [Aghion et al. \(2000, 2001\)](#) model with credit constraints, for instance, there is the possibility that an increase in interest rates could actually lead to exchange rate depreciations for a certain range of parameters. This possibility arises because higher domestic interest rates diminish investment and future output, reducing future demand for money – which implies a currency depreciation. On the other hand, [Blanchard \(2005\)](#), [Gonçalves and Guimarães \(2011\)](#) and [Hnatkovska et al. \(2011\)](#) develop fiscal dominance models in which high indebtedness levels can lead to an inverted relationship between interest and exchange rates. However these studies do not provide empirical evidence that interest hikes lead to significant increases in default or inflation risks.<sup>37</sup> The latter study, for instance, explains that “the higher interest rate raises the required seigniorage revenue to finance government spending and, *ceteris paribus*, increases the inflation rate (p. 3)” The authors do consider the possibility that interest rate hikes could lead to depreciation due to an output effect. However, in their calibration exercise this mechanism is shut down completely.

More recently, disaster risk models, along the lines of [Guo \(2007\)](#), Farhi and Gabaix (2008) and [Gourio et al. \(2013\)](#), have also delivered a negative association between interest rates and currency values, replicating the well-known failure of UIP. In these models, however, this is the result of a shock to disaster probability or to the resilience of a country, rather than to monetary policy itself.

To the extent that the exchange rate does not react to domestic monetary policy in the way that conventional theory and most standard open economy models would suggest, the results of this paper are also somewhat related to the exchange rate disconnect puzzle. [Moore and Roche \(2010\)](#) provide an explanation to the exchange rate disconnect puzzle and the failure of UIP based on habit persistence, in a paper that is closely related to [Verdelhan \(2010\)](#).

Finally, [Brunnermeier et al. \(2008\)](#) point out that carry trade would push the reaction of the exchange rate after an interest rate shock toward the one that is predicted by UIP. Slow adjustment could be due to liquidity constraints. However, their rationale suggests an initial under-reaction to a shock, rather than the reverse reaction that we find. The same holds for the infrequent portfolio adjustment theory of [Bacchetta and van Wincoop \(2010\)](#).

## 8. Concluding remarks

The failure to find the association that several authors have found in some developed countries leads us to conclude that in the context of emerging economies there is no indication that the exchange rate puzzle that is characteristic of the VAR literature is due to reverse causality or even fiscal dominance. When analyzing the time variation of  $\beta$  by using rolling window estimations we also did not find any evidence that the point estimates or significance of the coefficient changed in a noticeable way after the global financial crisis. In face of this, what we have in these countries is clearly a stronger version of the exchange rate puzzle.

It is apparent that the link between interest and exchange rate variations seems to be much more elusive than what the current generation of open economy models would suggest. Addressing this apparent failure of the exchange rate predictions of the standard interest parity cum rational expectations paradigm would certainly enhance, among others, the understanding of international capital flows. While recently some progress has been made in this respect, our view is that some papers have put far too great weight on fiscal dominance – without presenting the necessary evidence that interest

<sup>37</sup> In fact, we have shown the opposite in Section 4.

rate hikes are associated with increases in risk premia or inflation. The output effect of interest rate changes may have received too little weight in environments in which credit constraints tend to be important. The weakness of the domestic interest – exchange rate link also has implications for optimal monetary policy. What is clear is that finding possible solutions to the puzzle that was presented here should be placed high on the research agenda.

## Data sources

Exchange rates, interest rates, monetary policy committee meeting and exchange market intervention dates were obtained from the respective central banks, *Bloomberg* and *Thomson Reuters*. The 3 month T-Bill and the 1 month US Dollar LIBOR rates were obtained from the *Federal Reserve Bank of St. Louis*, whereas the VIX, CDS spreads and the CRB index were obtained through *Bloomberg*. Bilateral trade flows were taken from the IMF's Direction of Trade Statistics.

**Table A1**  
Testing for Non-Linearities based on Deviations

| D.V.: change in BRL/USD rate                                                          | Full sample |         | No intervention |        |
|---------------------------------------------------------------------------------------|-------------|---------|-----------------|--------|
|                                                                                       | L.S.        | L.S.    | L.S.            | L.S.   |
| Deviation based on interest rate differential                                         |             |         |                 |        |
| d (DI 90)                                                                             | 1.028       | 1.330   | 1.600           | 1.564  |
|                                                                                       | 0.892       | 0.886   | 1.409           | 1.346  |
| d (DI 90)* dummy (deviation >90th pctl)                                               | –0.584      | 0.547   | –1.378          | –0.038 |
|                                                                                       | 1.194       | 1.247   | 1.605           | 1.774  |
| Control variables                                                                     | No          | Yes     | No              | Yes    |
| No. of observations                                                                   | 79          | 79      | 38              | 38     |
| R2                                                                                    | 0.025       | 0.211   | 0.073           | 0.300  |
| RMSE                                                                                  | 1.009       | 0.926   | 0.952           | 0.866  |
| F/chi2                                                                                | 0.92        | 3.55*** | 0.69            | 2.83** |
| Deviation based on interest rate differential minus expected depreciation             |             |         |                 |        |
| d (DI 90)                                                                             | 0.783       | 1.059   | 1.746           | 1.538  |
|                                                                                       | 0.912       | 0.890   | 1.398           | 1.312  |
| d (DI 90)* dummy (deviation >90th pctl)                                               | 1.293       | 2.473   | –2.922          | 0.236  |
|                                                                                       | 1.754       | 1.539   | 1.937           | 1.860  |
| Control variables                                                                     | No          | Yes     | No              | Yes    |
| No. of observations                                                                   | 79          | 79      | 38              | 38     |
| R2                                                                                    | 0.030       | 0.228   | 0.094           | 0.299  |
| RMSE                                                                                  | 1.006       | 0.916   | 0.942           | 0.865  |
| F/chi2                                                                                | 1.32        | 6.23*** | 1.16            | 2.82** |
| Deviation based on interest rate differential (risk adjusted)                         |             |         |                 |        |
| d (DI 90)                                                                             | 1.021       | 1.332   | 1.591           | 1.579  |
|                                                                                       | 0.894       | 0.888   | 1.409           | 1.347  |
| d (DI 90)* dummy (deviation >90th pctl)                                               | –0.476      | 0.517   | –1.301          | –0.174 |
|                                                                                       | 1.229       | 1.238   | 1.635           | 1.745  |
| Control variables                                                                     | No          | Yes     | No              | Yes    |
| No. of observations                                                                   | 79          | 79      | 38              | 38     |
| R2                                                                                    | 0.025       | 0.210   | 0.072           | 0.300  |
| RMSE                                                                                  | 1.009       | 0.926   | 0.952           | 0.866  |
| F/chi2                                                                                | 0.99        | 3.58*** | 0.71            | 2.84** |
| Deviation based on interest rate differential minus expected depreciation (risk adj.) |             |         |                 |        |
| d (DI 90)                                                                             | 0.674       | 1.073   | 1.587           | 1.584  |
|                                                                                       | 0.911       | 0.893   | 1.385           | 1.328  |
| d (DI 90)* dummy (deviation>90th pctl)                                                | 2.044       | 2.333   | –1.470          | –0.243 |
|                                                                                       | 1.521       | 1.658   | 1.548           | 1.668  |
| Control variables                                                                     | no          | yes     | no              | yes    |
| No. of observations                                                                   | 79          | 79      | 38              | 38     |
| R2                                                                                    | 0.038       | 0.225   | 0.073           | 0.300  |
| RMSE                                                                                  | 1.002       | 0.918   | 0.952           | 0.866  |
| F/chi2                                                                                | 2.72*       | 5.34*** | 0.68            | 2.84** |

Note: Robust standard errors. \*, \*\* and \*\*\* denote statistical significance at the 10%, 5% and 1% confidence levels, respectively.

**Table A2**

The Impact of Monetary Policy on the Exchange Rate

| D.V.: change in BRL/USD rate | Full sample |          | No intervention |         |
|------------------------------|-------------|----------|-----------------|---------|
|                              | L.S.        | L.S.     | L.S.            | L.S.    |
| Brazil                       |             |          |                 |         |
| SELIC – E (SELIC)            | 0.391       | 0.238    | 0.544           | 0.400   |
|                              | 0.358       | 0.307    | 0.440           | 0.317   |
| d (3 month T Bill)           |             | 1.116    |                 | 3.515   |
|                              |             | 2.931    |                 | 3.107   |
| d (VIX)                      |             | 0.283*** |                 | 0.225** |
|                              |             | 0.081    |                 | 0.097   |
| d (price of commodities)     |             | 0.083    |                 | 0.328   |
|                              |             | 0.245    |                 | 0.278   |
| No. of observations          | 78          | 78       | 37              | 37      |
| R2                           | 0.008       | 0.150    | 0.031           | 0.194   |
| RMSE                         | 0.977       | 0.923    | 0.869           | 0.829   |
| F/chi2                       | 1.20        | 3.87***  | 1.53            | 7.15*** |
| Durbin score (chi 2)         | 1.298       | 0.101    | 0.012           | 1.138   |
| Wu–Hausman (F)               | 1.303       | 0.093    | 0.011           | 1.165   |

Note: Robust standard errors. \*, \*\* and \*\*\* denote statistical significance at the 10%, 5% and 1% confidence levels, respectively.

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